Model Analysis

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1 Data analysis /Dataframe import and filters

1.1 DataFrame import and filters

The data has been imported with the following(s) function(s):

```
def build_df():
    if flag_IR:
        df, features_engineered = import_data_IR()
        if flag_IR_maskP:
            df = mask_period_IR(df)
        if flag_IR_balR:
            df = mask_balanceHC_random_IR(df, n_samples=None)
    if flag_ER:
        df, features_engineered = import_data_ER()

    cols_to_check = target_col + features_X
    mask = np.isfinite(df[cols_to_check]).all(axis=1)
    df = df[mask]


    cols = list(dict.fromkeys(target_col + features_X  +
                              feats_to_balance + feats_grouping +
                              feats_reference+ feats_postproc))
    df = df[cols]
```

```python
    #df['campaign'] = "ALL"

    return df
```

```python
def import_data_IR():
    if target_col[0] == "i_NH3":
        #file_path = "csv/innova/switched/df_signal_decomposition_NH3_current_db2.csv"
        file_path = "csv/innova/notswitched/df_signal_decomposition_current.csv"
    elif target_col[0] == "i_CO2":
        file_path = "csv/innova/switched/df_signal_decomposition_CO2_current.csv"
    else:
        raise ValueError("Target is not supported")

    df = pd.read_csv(file_path, sep=";")
    df["datetime"] = pd.to_datetime(df["datetime"])
    df.sort_values("datetime").reset_index(drop=True)

    if target_col[0] == "i_NH3":
        df["campaign"] = df["datetime"].apply(assign_campaign)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/notswitched/features_decomposition_list.json", "r") as f:
            dict_features = json.load(f)
    elif target_col[0] == "i_CO2":
        df["campaign"] = df["datetime"].apply(assign_campaign_iCO2)
        df = df[df["campaign"] != "other"].copy()
        df["innova_point"] = df.apply(assign_innova_point_iCO2, axis=1)
        df = df[df["innova_point"] != "no"].copy()
        with open("csv/innova/switched/features_decomposition_CO2_list.json", "r") as f:
            dict_features = json.load(f)
    else:
        raise ValueError("The target col do not match any existing function")


    df["hot"] = df.apply(assign_hot_season,axis=1)
    df["umidity_range"] = df.apply(assign_humidity, axis = 1)
    #Removing of rabbit 4
    df["column"] = df.apply(assign_colonna, axis = 1)
    df['height'] = df.apply(assign_altezza, axis = 1)
```

```
    return df,dict_features
```

```
def assign_innova_point(row):
    camp = row["campaign"]
    rid = row["id_rabbit"]

    # if camp == "2025_Jun":
    #     mapping = {2: "sl11", 4: "sl5", 6: "sl4", 1: "sl3", 5: "no"}
    # elif camp == "2025_Sep":
    #     mapping = {2: "sl11", 4: "sl5", 3: "sl4", 6: "sl3", 5: "no"}
    if camp == "2025_Apr":
        #mapping = {1: "no", 2: "no", 4: "no", 6: "sl5"}
        #mapping = {1: "no", 2: "sl5", 4: "no", 6: "sl5"}
        mapping = {1: "no", 2: "no", 4: "no", 6: "no"}
    elif camp == "2025_Jun":
        mapping = {2: "sl11", 4: "sl5", 1: "sl3", 6: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 1: "sl4", 6: "sl3", 5: "no"}
    elif camp == "2025_Sep":
        mapping = {2: "sl11", 4: "sl5", 6: "sl3", 3: "sl4", 5: "no"}
        #mapping = {2: "sl11", 4: "no", 6: "sl4", 3: "sl3", 5: "no"}
    elif camp == "2026_Gen":
        #mapping = {1: "no", 4: "no", 5: "sl11", 6: "no"}
        #mapping = {1: "sl11", 4: "no", 5: "sl11", 6: "sl11"}
        mapping = {1: "no", 4: "no", 5: "no", 6: "no"}
    else:
        mapping = {}

    return mapping.get(rid, "no")
```

Function print :

1.2 Metadata

Nombre de lignes df	8168
Nombre de lignes df clean	8168
Nombre de features	5
% de NaN	0.0
Taille dataset entraînement	5717
Taille dataset test	2450
Métrique utilisée	r2

Column target : i_NH3

Features_X : r_NH3, r_temperature, r_umidity, r_CO2, r_THI

feats_to_balance :

feats_grouping :

feats_postproc : datetime, id_channel, id_rabbit, campaign

feats_ref :

id_cols : campaign

2 Model(s) analysis

2.1 Explanation legend tables

Columns

- rank: ranking regarding cross validation test
- train_cv: mean score train of the CV
- test_cv: mean score test of the CV
- test: score on the testing dataset
- diff: test-test_cv
- r_test: test/test_cv
- r_train: train_cv/test_cv

Colors:

[Orange] r_test < 0.95 (under performance of the test)

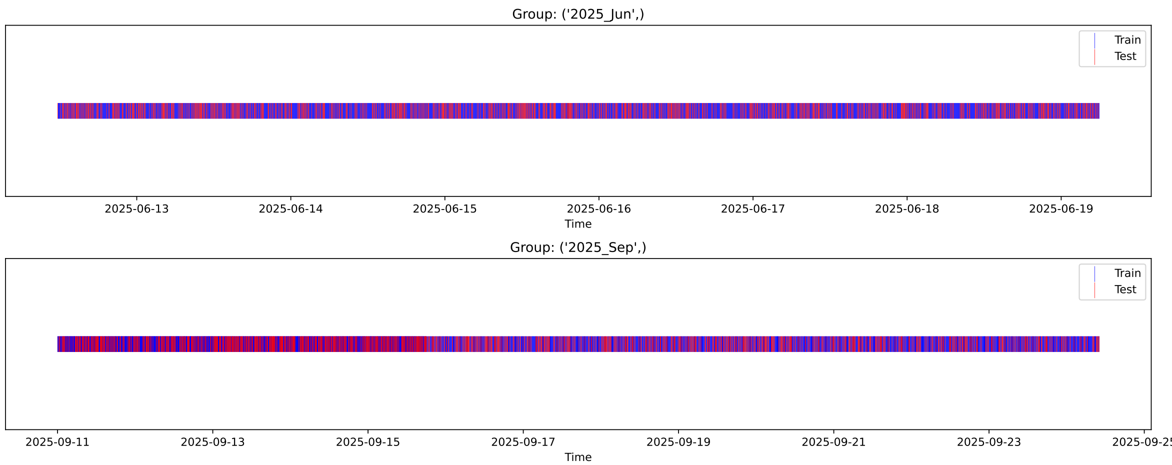
[Green] r_test > 1.05 (over performance of the test)

[Red] r_train not in [0.95, 1.05] (gap between train/test)

2.2 Repartition of train and test

2.2.1 External Split: X_train vs X_test

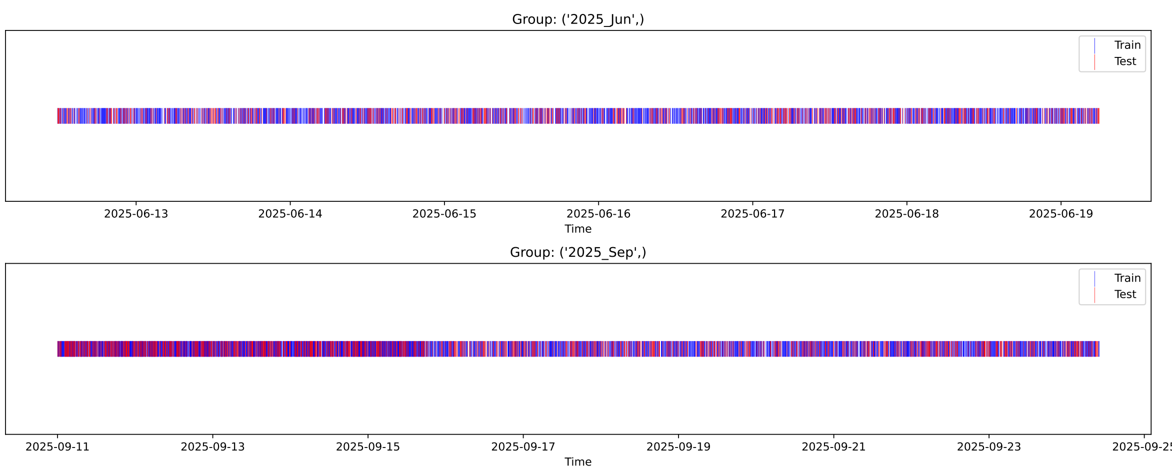
External Split: Train: 5717 samples Test: 2451 samples



2.2.2 Internal CV Folds

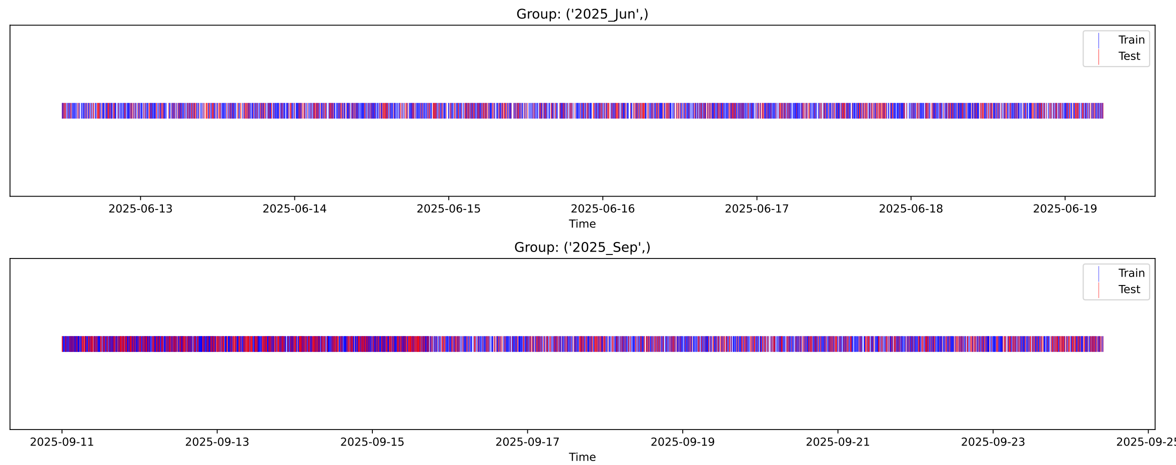
Fold 1/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



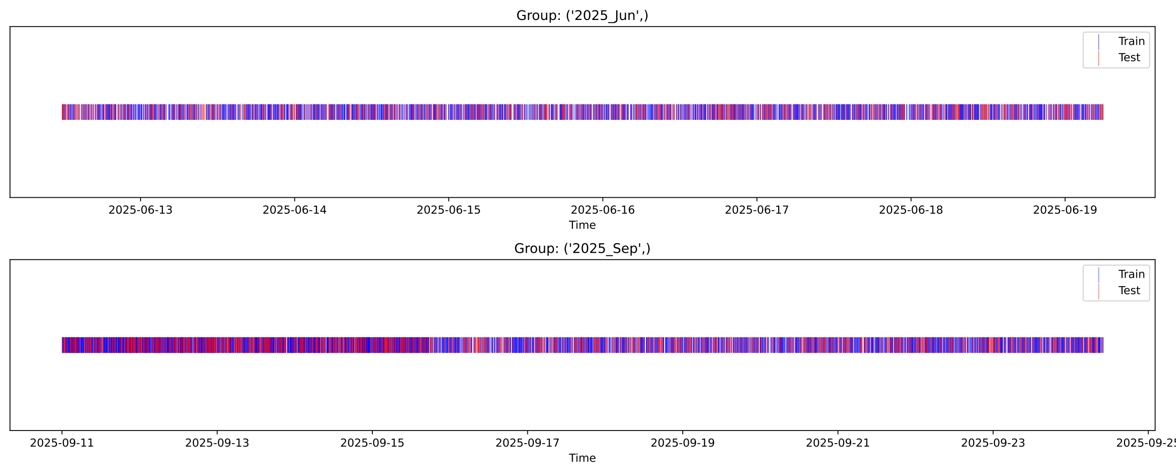
Fold 2/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



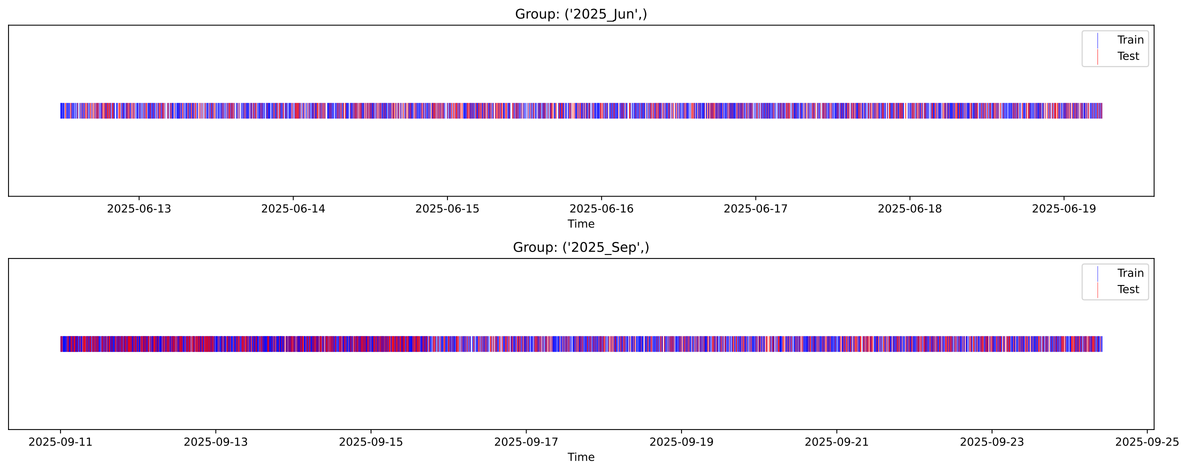
Fold 3/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



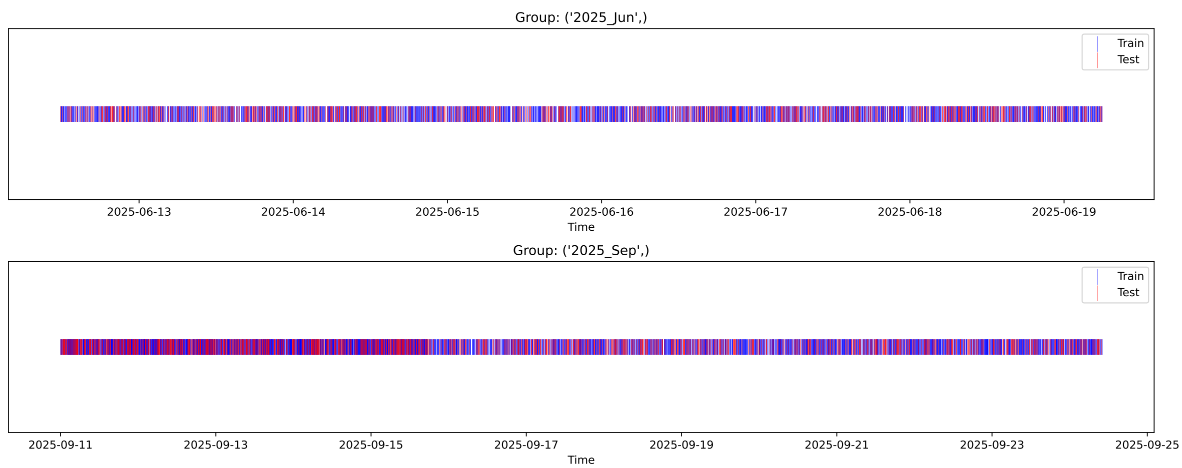
Fold 4/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



Fold 5/5

Train: 4001 samples Test: 1716 samples Excluded: 0 samples Window exclusion: 0h



2.3 Model : Linear

2.3.1 Gridsearch

Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

Fold 0 — y_test: mean=2.34, std=0.82, y_train: mean=2.34, std=0.79

Fold 1 — y_test: mean=2.34, std=0.77, y_train: mean=2.34, std=0.81

Fold 2 — y_test: mean=2.36, std=0.85, y_train: mean=2.33, std=0.78

Fold 3 — y_test: mean=2.36, std=0.84, y_train: mean=2.33, std=0.78

Fold 4 — y_test: mean=2.35, std=0.76, y_train: mean=2.34, std=0.81

train scores CV: [0.35126041 0.34129782 0.38941971 0.36572649 0.35147845]

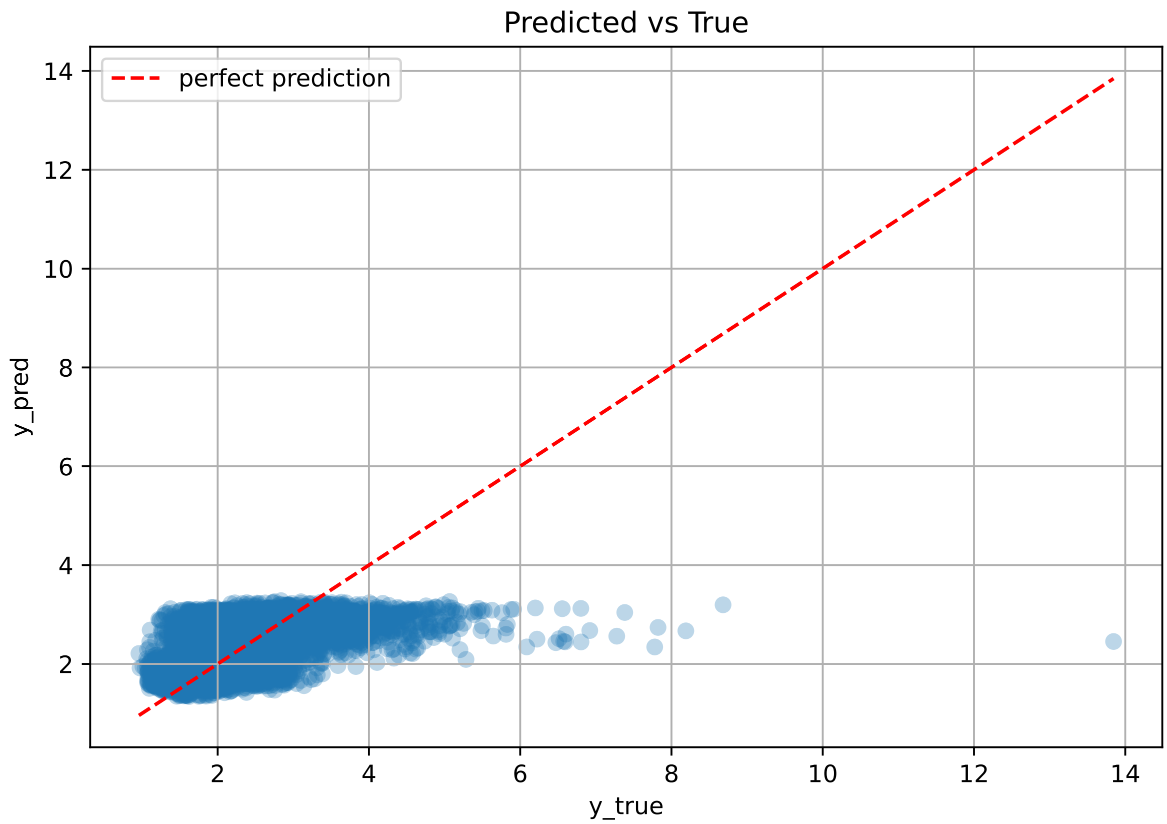
test scores CV: [0.37881413 0.40998904 0.30281638 0.34887505 0.38406497]

rank	mean_train_cv	mean_test_cv	std_test_cv	test_score_cv	test	diff	r_test	r_train
1.0	0.3598	0.3649	0.0366	0.3451	0.3453	-0.0197	0.9461	0.9861

Coefficients:

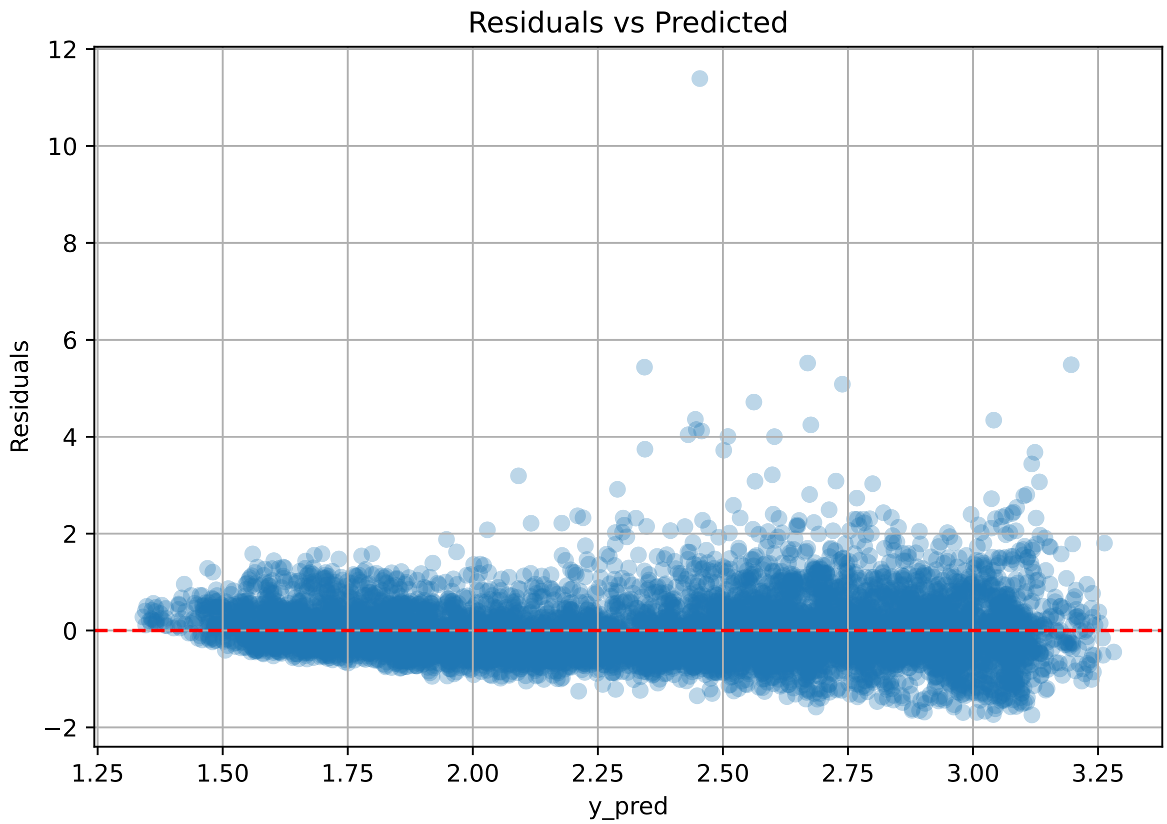
coef_r_CO2 = -0.0491, coef_r_NH3 = -0.0447, coef_r_THI = 0.5970, coef_r_temperature = -0.7372, coef_r_umidity = -0.6671, intercept = 2.3392,

2.3.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
2
4
6
8
10
12
14

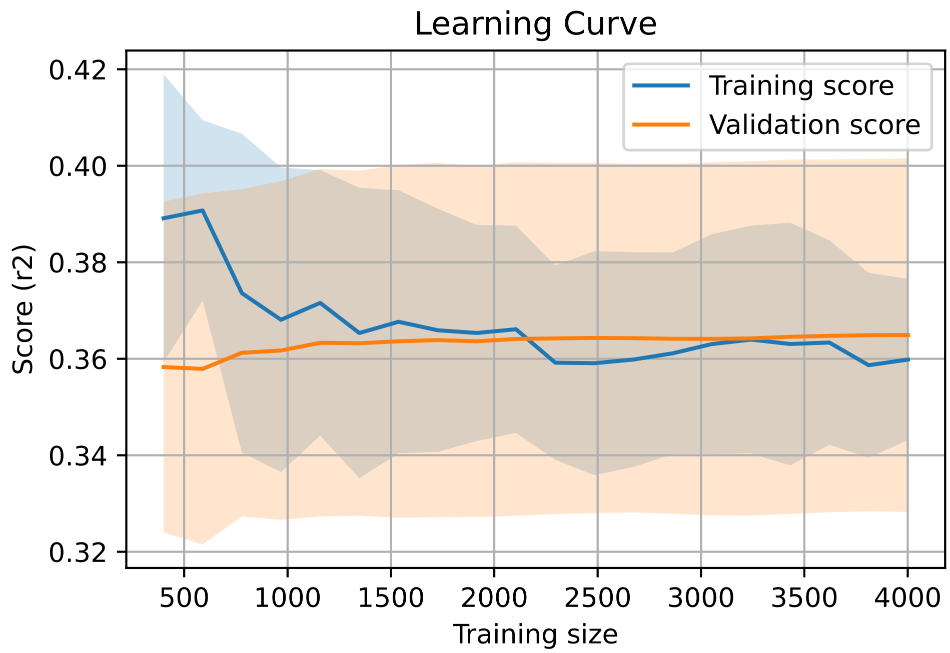
2.3.3 Scatter residuals



2.3.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

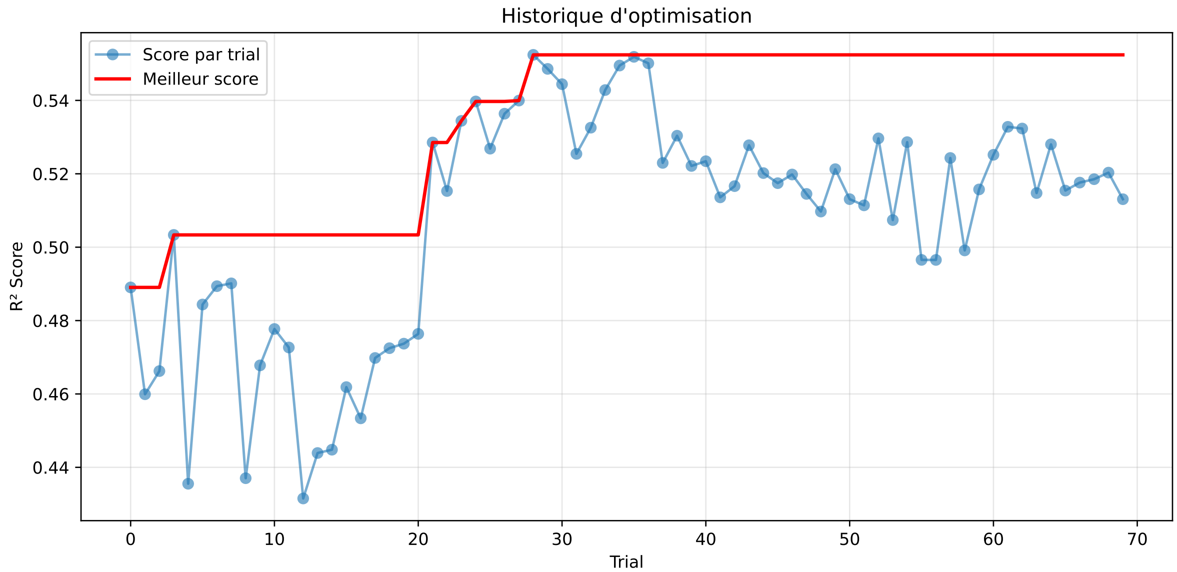


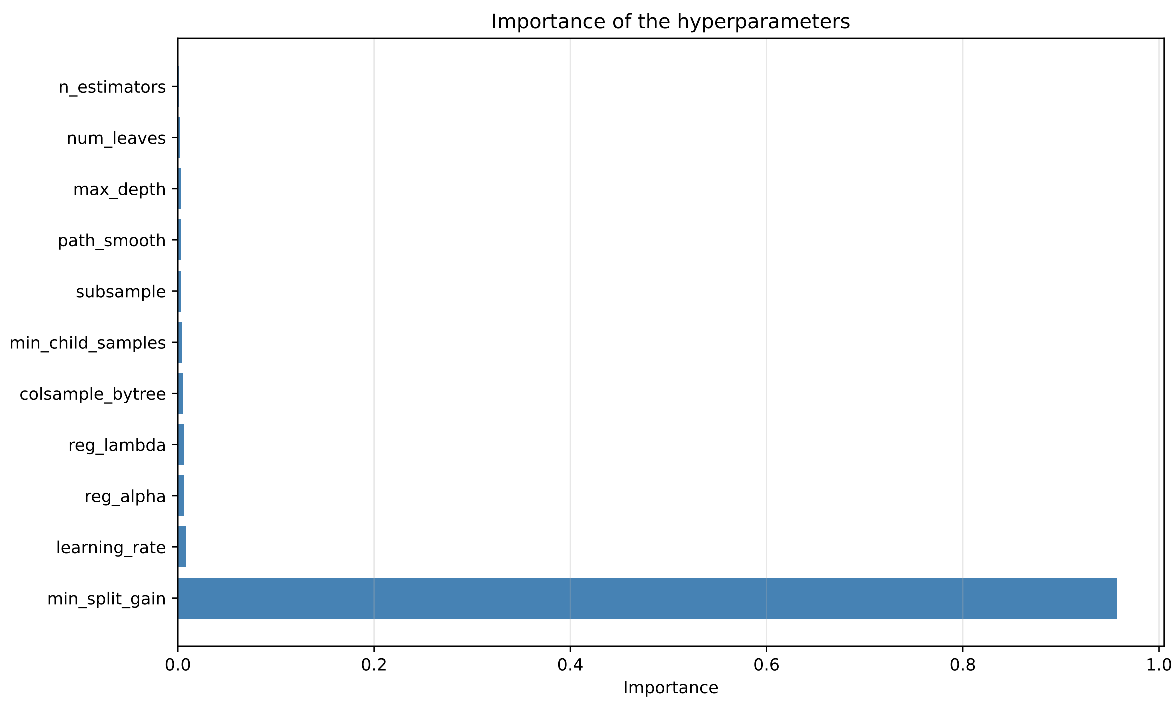
2.4 Model : LGBM

2.4.1 Gridsearch

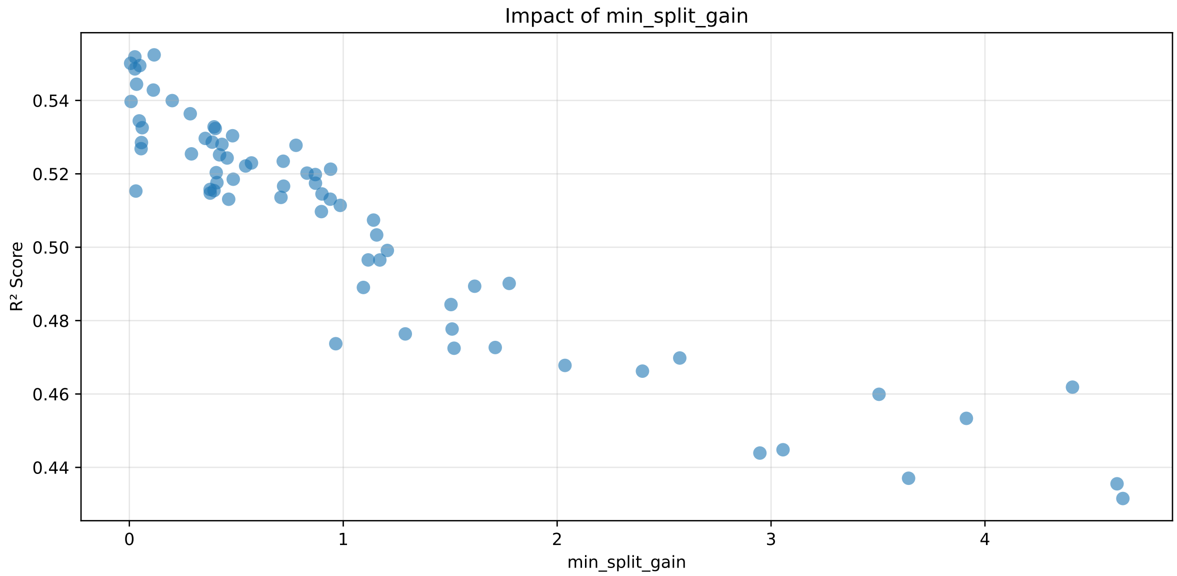
Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

==================== TOP Importance FEATURES ==================== feature importance r_umidity 252 r_CO2 239 r_NH3 167 r_THI 124 r_temperature 105





Paramètres avec >5% d'importance : ['min_split_gain']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.4643	0.4439	0.4542	0.037	0.4439	0.4799	0.0359	1.081	1.0459
2	0.4575	0.437	0.4466	0.0339	0.437	0.4781	0.0411	1.094	1.0468
3	0.4561	0.4355	0.4463	0.0352	0.4355	0.4754	0.0398	1.0915	1.0471
4	0.4486	0.4315	0.4407	0.0376	0.4315	0.4665	0.035	1.0812	1.0397
5	0.5067	0.4777	0.497	0.0342	0.3909	0.5078	0.0301	1.063	1.0606
6	0.5063	0.4764	0.4964	0.0359	0.3866	0.5154	0.039	1.0819	1.0628
7	0.5029	0.4737	0.4901	0.0353	0.386	0.506	0.0323	1.0682	1.0617
8	0.4876	0.4599	0.4738	0.0365	0.3767	0.4972	0.0373	1.0811	1.0603
9	0.4795	0.4534	0.4685	0.0375	0.375	0.4957	0.0423	1.0934	1.0576
10	0.4684	0.4448	0.4575	0.0369	0.3741	0.4923	0.0475	1.1069	1.053
---	---	---	---	---	---	---	---	---	---
70	0.6983	0.5519	0.5856	0.036	0.1125	0.5932	0.0413	1.0749	1.2654

Hyperparameters:

Rank 1: {'reg___n_estimators': 400, 'reg___max_depth': 3, 'reg___learning_rate': 0.026575098856952546, 'reg___num_leaves': 23, 'reg___subsample': 0.6175610989774247, 'reg___colsample_bytree': 0.6012441677659953, 'reg___min_child_samples': 29, 'reg___reg_alpha': 6.984396449164003, 'reg___reg_lambda': 29.131561107077953, 'reg___min_split_gain': 2.9478362301059264, 'reg___path_smooth': 3.3136326809079257}

Rank 2: {'reg___n_estimators': 800, 'reg___max_depth': 5, 'reg___learning_rate': 0.04592449105251225, 'reg___num_leaves': 32, 'reg___subsample': 0.7669131243639693, 'reg__colsample_bytree': 0.6089589057577491, 'reg__min_child_samples': 44, 'reg__reg_alpha': 5.83041107046776, 'reg__reg_lambda': 49.00300338776019, 'reg__min_split_gain': 3.6429108407613056, 'reg__path_smooth': 3.6192959433567093}

Rank 3: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.011432269926351978, 'reg__num_leaves': 29, 'reg__subsample': 0.8851718063301884, 'reg__colsample_bytree': 0.6566064485036357, 'reg__min_child_samples': 31, 'reg__reg_alpha': 7.3705482467686965, 'reg__reg_lambda': 17.04408694033841, 'reg__min_split_gain': 4.617511459565826, 'reg__path_smooth': 1.3792861869308959}

Rank 4: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.012378984442051669, 'reg__num_leaves': 20, 'reg__subsample': 0.5124396720284267, 'reg__colsample_bytree': 0.7598149317639906, 'reg__min_child_samples': 25, 'reg__reg_alpha': 5.718197572409327, 'reg__reg_lambda': 44.72321450158676, 'reg__min_split_gain': 4.644802765234014, 'reg__path_smooth': 1.0739025547853402}

Rank 5: {'reg__n_estimators': 500, 'reg__max_depth': 3, 'reg__learning_rate': 0.010606560938712262, 'reg__num_leaves': 29, 'reg__subsample': 0.7578904478492678, 'reg__colsample_bytree': 0.6441649099186081, 'reg__min_child_samples': 40, 'reg__reg_alpha': 2.5641062310335916, 'reg__reg_lambda': 19.824059666419892, 'reg__min_split_gain': 1.5089478866940405, 'reg__path_smooth': 0.3774163370058453}

Rank 6: {'reg__n_estimators': 600, 'reg__max_depth': 3, 'reg__learning_rate': 0.014127773834351343, 'reg__num_leaves': 61, 'reg__subsample': 0.6701745952201872, 'reg__colsample_bytree': 0.841548040698169, 'reg__min_child_samples': 38, 'reg__reg_alpha': 5.562884505009153, 'reg__reg_lambda': 29.715113323457594, 'reg__min_split_gain': 1.2900887887616501, 'reg__path_smooth': 1.3574654151323502}

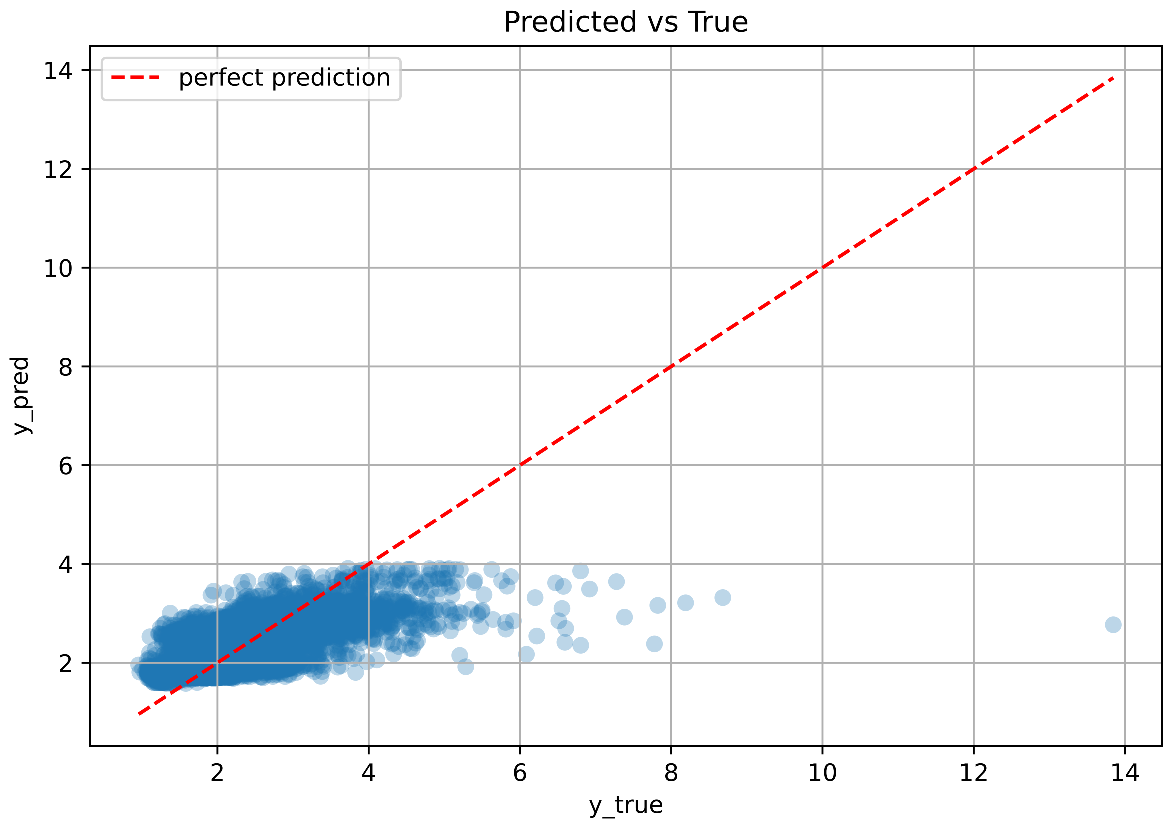
Rank 7: {'reg__n_estimators': 400, 'reg__max_depth': 3, 'reg__learning_rate': 0.06471236293376403, 'reg__num_leaves': 43, 'reg__subsample': 0.8095456055424233, 'reg__colsample_bytree': 0.5841801221434496, 'reg__min_child_samples': 15, 'reg__reg_alpha': 8.673298972627649, 'reg__reg_lambda': 31.382390899447525, 'reg__min_split_gain': 0.9650976533964334, 'reg__path_smooth': 0.6776773259747493}

Rank 8: {'reg__n_estimators': 600, 'reg__max_depth': 6, 'reg__learning_rate': 0.06562449687755963, 'reg__num_leaves': 50, 'reg__subsample': 0.847935441558095, 'reg__colsample_bytree': 0.637393669594288, 'reg__min_child_samples': 18, 'reg__reg_alpha': 1.9878124030256683, 'reg__reg_lambda': 32.0313263878656, 'reg__min_split_gain': 3.5047646022094754, 'reg__path_smooth': 0.13421398191218403}

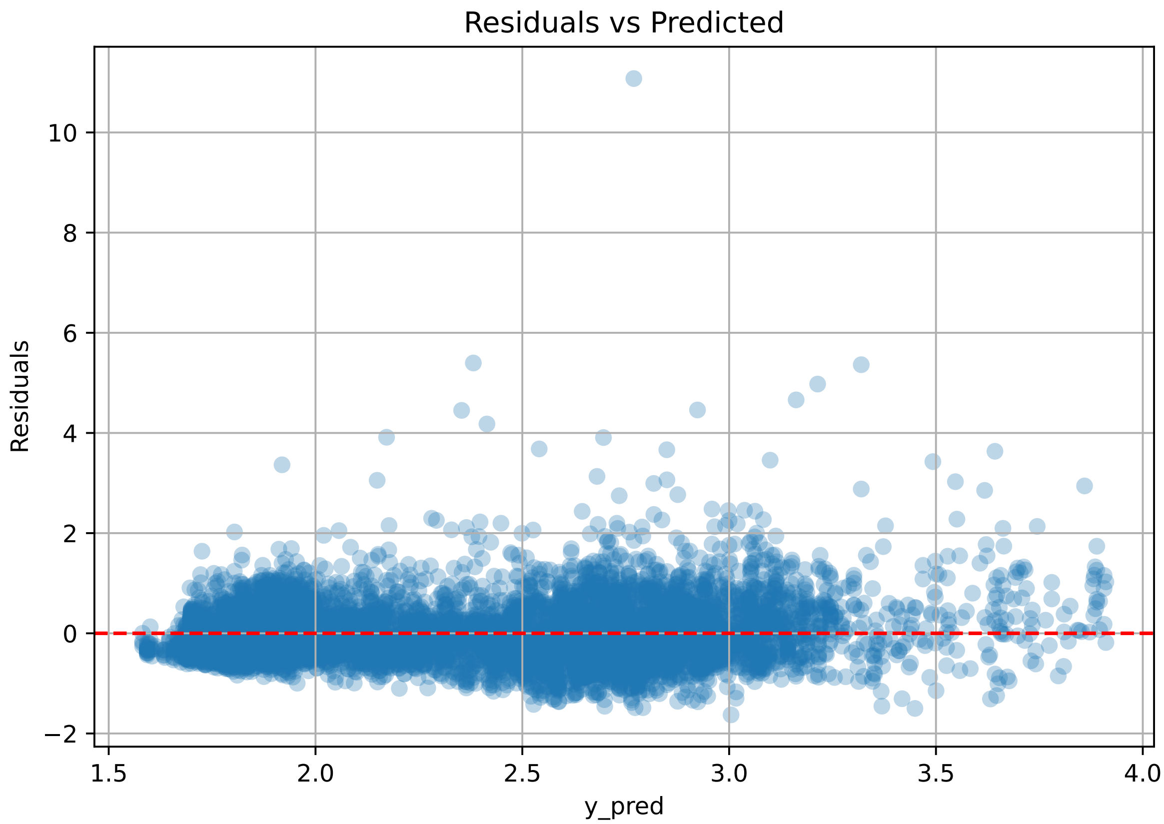
Rank 9: {'reg__n_estimators': 200, 'reg__max_depth': 5, 'reg__learning_rate': 0.06077416569929432, 'reg__num_leaves': 53, 'reg__subsample': 0.7548747076511126, 'reg__colsample_bytree': 0.777174982593303, 'reg__min_child_samples': 24, 'reg__reg_alpha': 6.479551056831153, 'reg__reg_lambda': 7.649058677853682, 'reg__min_split_gain': 3.9132406710334906, 'reg__path_smooth': 4.597863260043745}

Rank 10: {'reg___n_estimators': 500, 'reg___max_depth': 7, 'reg___learning_rate': 0.010430152600043683, 'reg___num_leaves': 28, 'reg___subsample': 0.8309105701450417, 'reg___colsample_bytree': 0.8884076659810266, 'reg___min_child_samples': 11, 'reg___reg_alpha': 9.549445335679124, 'reg___reg_lambda': 43.35171706019457, 'reg___min_split_gain': 3.0559784594720174, 'reg___path_smooth': 1.2712808654165815}

2.4.2 Scatter



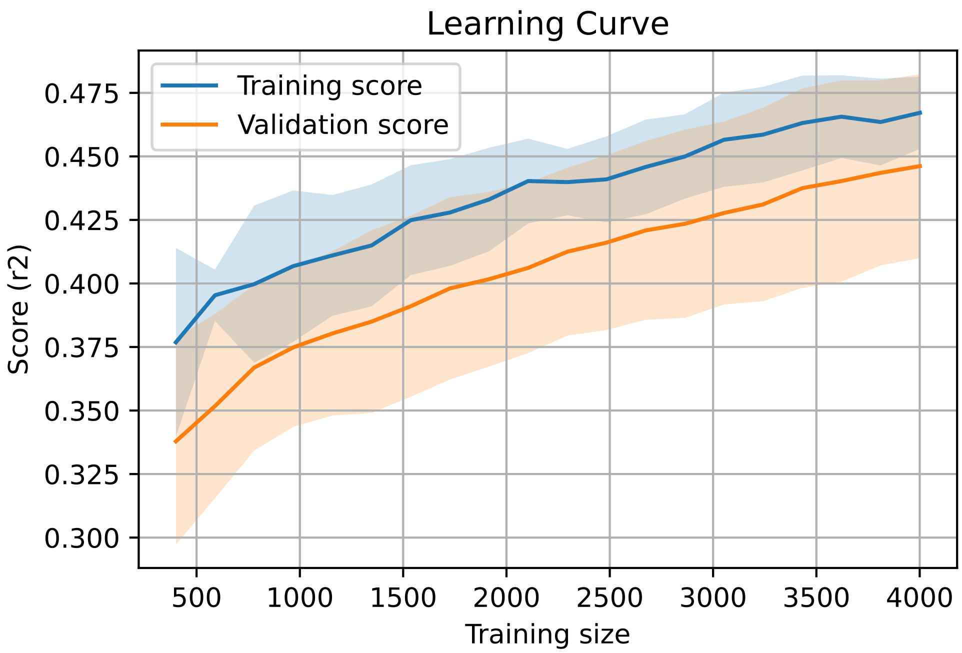
2.4.3 Scatter residuals



2.4.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450

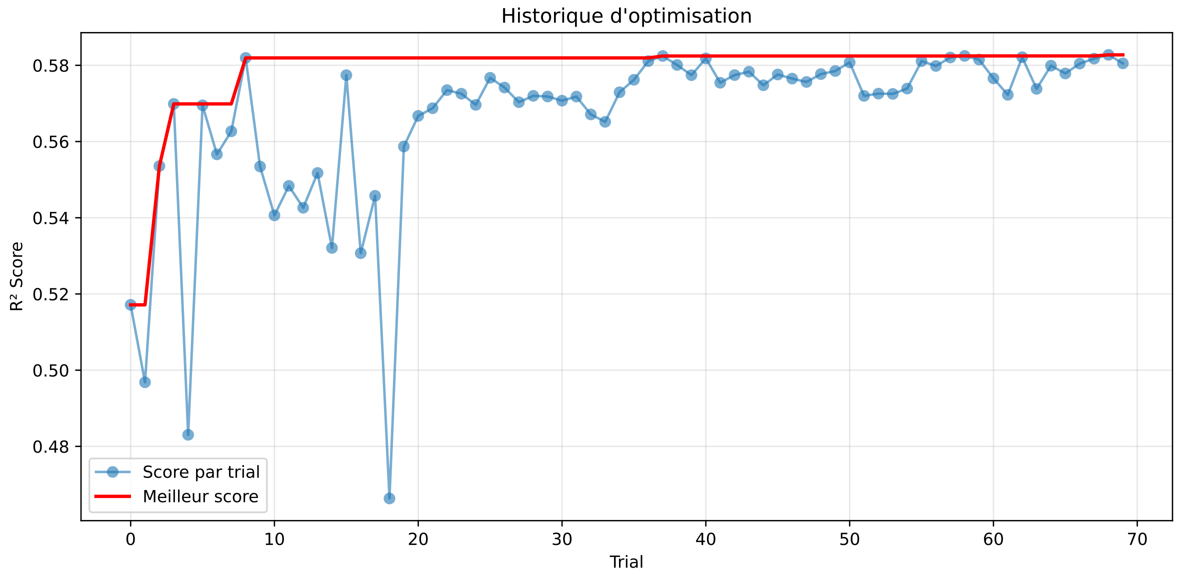


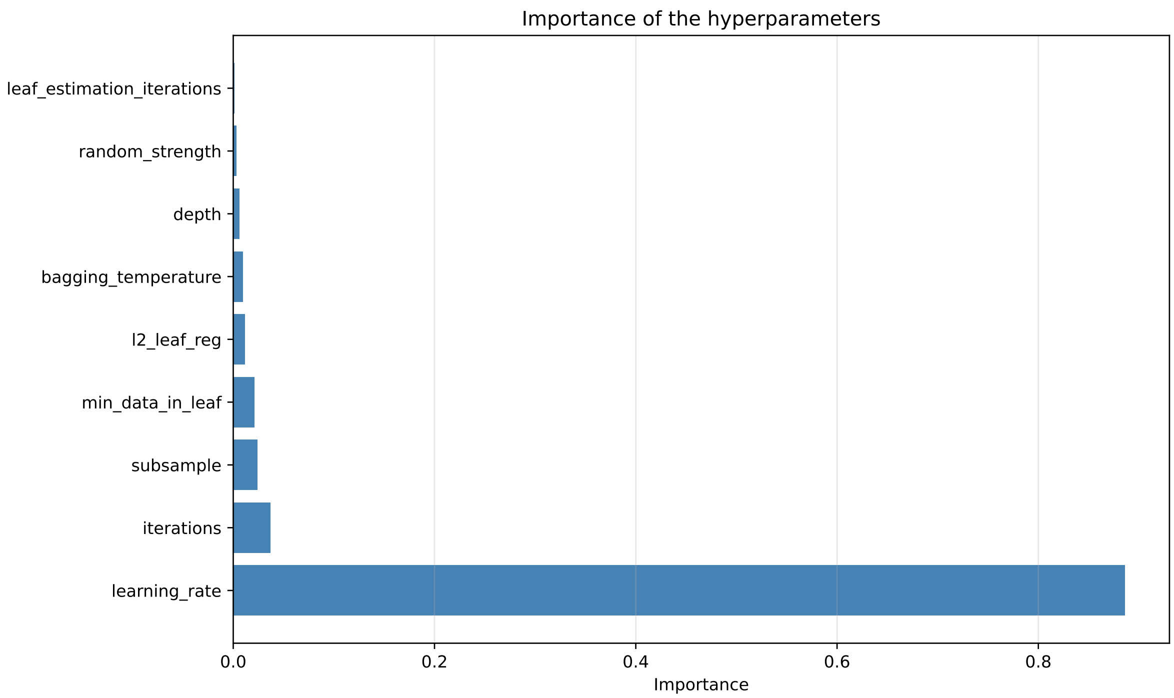
2.5 Model : CatBoost

2.5.1 Gridsearch

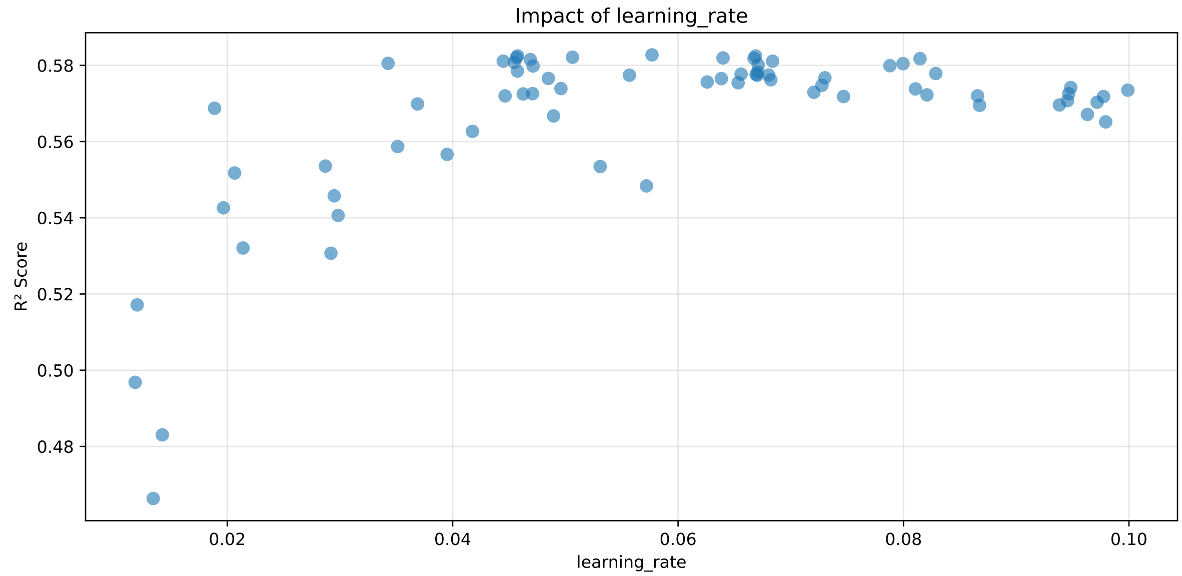
Distribution y_train vs y_test: y_train: mean=2.339, std=0.798, min=0.959, max=13.845 y_test: mean=2.374, std=0.786, min=1.071, max=7.277

==================== TOP Importance FEATURES ==================== feature importance r_umidity 42.582236 r_NH3 20.944685 r_CO2 17.530528 r_temperature 9.478563 r_THI 9.463989





Paramètres avec >5% d'importance : ['learning_rate']



Pas assez de paramètres avec >5% d'importance pour une heatmap 2D.

rank	mean_train_cv	mean_test_cv	test_score_cv	std_test_cv	penalized_score	test	diff	r_test	r_train
1	0.5056	0.483	0.5013	0.0368	0.483	0.5064	0.0234	1.0484	1.0467
2	0.4837	0.4663	0.4819	0.0378	0.4663	0.4879	0.0215	1.0462	1.0371
3	0.5243	0.4968	0.5193	0.0378	0.4144	0.5264	0.0296	1.0595	1.0553
4	0.5772	0.5307	0.5588	0.0378	0.3912	0.5648	0.0342	1.0644	1.0876
5	0.5591	0.5171	0.5421	0.0374	0.3911	0.5485	0.0314	1.0607	1.0812
6	0.5945	0.5406	0.5596	0.0351	0.3789	0.5635	0.0229	1.0423	1.0997
7	0.6021	0.5426	0.5663	0.0371	0.3642	0.5709	0.0283	1.0522	1.1096
8	0.5952	0.5321	0.5598	0.0366	0.3428	0.5676	0.0355	1.0668	1.1186
9	0.6198	0.5484	0.5735	0.034	0.3341	0.5789	0.0305	1.0556	1.1302
10	0.627	0.5536	0.5772	0.0364	0.3332	0.5863	0.0327	1.0591	1.1327
---	---	---	---	---	---	---	---	---	---
70	0.9443	0.5652	0.6042	0.0363	-0.5722	0.62	0.0548	1.097	1.6708

Hyperparameters:

Rank 1: {'reg___iterations': 300, 'reg___depth': 5, 'reg___learning_rate': 0.014240929798738535, 'reg___l2_leaf_reg': 17.91725243282948, 'reg___bagging_temperature': 4.210939966207815, 'reg___random_strength': 2.6860402200247635, 'reg___subsample': 0.8719128150151771, 'reg___min_data_in_leaf': 20, 'reg___leaf_estimation_iterations': 4}

Rank 2: {'reg___iterations': 400, 'reg___depth': 3, 'reg___learning_rate': 0.013432461946559919, 'reg___l2_leaf_reg': 40.23817002592688, 'reg___bagging_temperature': 1.1642801660318853, 'reg___random_strength': 1.060842641710538, 'reg___subsample': 0.564311097953264, 'reg___min_data_in_leaf': 32, 'reg___leaf_estimation_iterations': 3}

Rank 3: {'reg__iterations': 400, 'reg__depth': 6, 'reg__learning_rate': 0.01183269789865813, 'reg__l2_leaf_reg': 32.67089814894085, 'reg__bagging_temperature': 0.17100432478318328, 'reg__random_strength': 3.5555612334767703, 'reg__subsample': 0.8465667954130562, 'reg__min_data_in_leaf': 38, 'reg__leaf_estimation_iterations': 6}

Rank 4: {'reg__iterations': 300, 'reg__depth': 6, 'reg__learning_rate': 0.029199220439139962, 'reg__l2_leaf_reg': 17.931349997392296, 'reg__bagging_temperature': 2.7134005221199473, 'reg__random_strength': 3.7712341816998594, 'reg__subsample': 0.7590013794907131, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 4}

Rank 5: {'reg__iterations': 400, 'reg__depth': 7, 'reg__learning_rate': 0.012009858079869326, 'reg__l2_leaf_reg': 18.36121813483839, 'reg__bagging_temperature': 2.083342477268525, 'reg__random_strength': 3.037540739739948, 'reg__subsample': 0.5091349511715163, 'reg__min_data_in_leaf': 33, 'reg__leaf_estimation_iterations': 9}

Rank 6: {'reg__iterations': 700, 'reg__depth': 3, 'reg__learning_rate': 0.029834775749594138, 'reg__l2_leaf_reg': 17.77938734857729, 'reg__bagging_temperature': 2.6045240088099475, 'reg__random_strength': 1.966896986200661, 'reg__subsample': 0.6981572960471392, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 6}

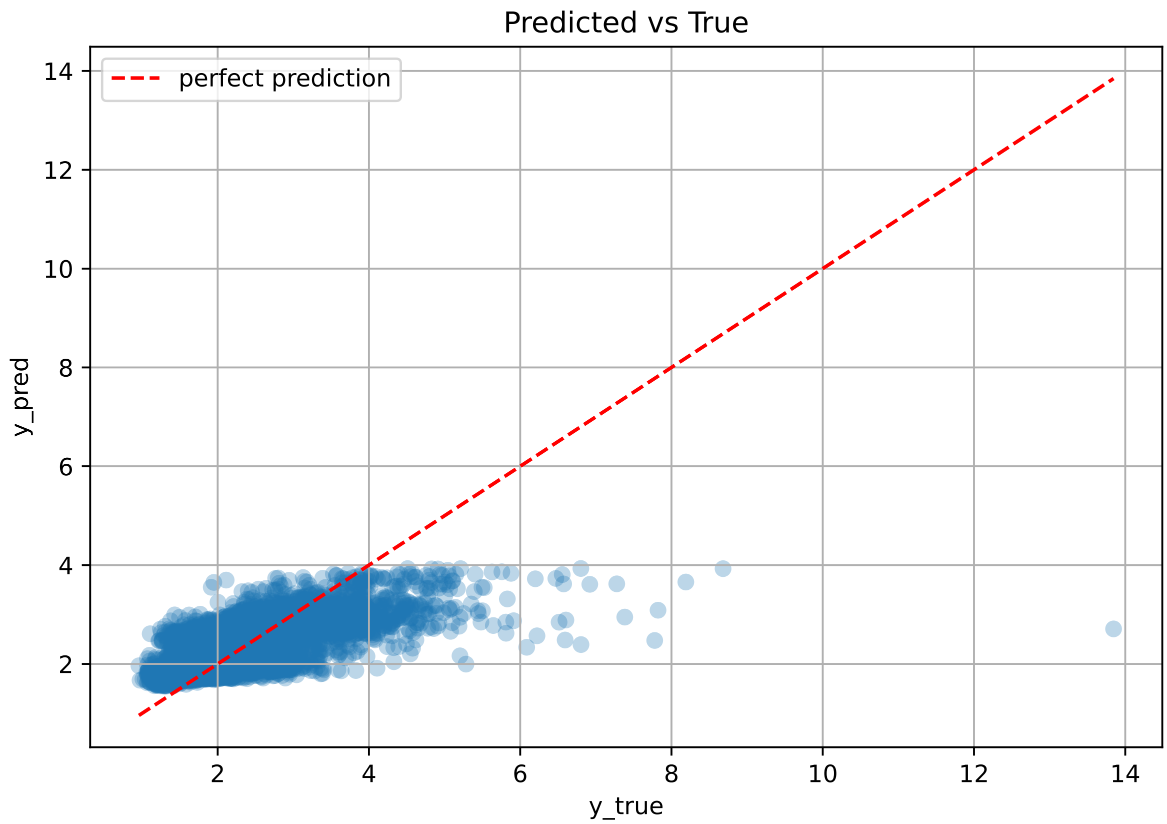
Rank 7: {'reg__iterations': 600, 'reg__depth': 5, 'reg__learning_rate': 0.019668418943123142, 'reg__l2_leaf_reg': 3.627028822896722, 'reg__bagging_temperature': 1.2836385979751208, 'reg__random_strength': 3.963699515103828, 'reg__subsample': 0.5817280644418263, 'reg__min_data_in_leaf': 20, 'reg__leaf_estimation_iterations': 4}

Rank 8: {'reg__iterations': 200, 'reg__depth': 8, 'reg__learning_rate': 0.021404726840766917, 'reg__l2_leaf_reg': 11.127104375921745, 'reg__bagging_temperature': 3.378498870172031, 'reg__random_strength': 1.4209118261398839, 'reg__subsample': 0.8237829976200476, 'reg__min_data_in_leaf': 12, 'reg__leaf_estimation_iterations': 5}

Rank 9: {'reg__iterations': 300, 'reg__depth': 4, 'reg__learning_rate': 0.05719575149392753, 'reg__l2_leaf_reg': 25.64597177864674, 'reg__bagging_temperature': 0.19343623086101602, 'reg__random_strength': 1.6283924494522966, 'reg__subsample': 0.773748996822706, 'reg__min_data_in_leaf': 31, 'reg__leaf_estimation_iterations': 6}

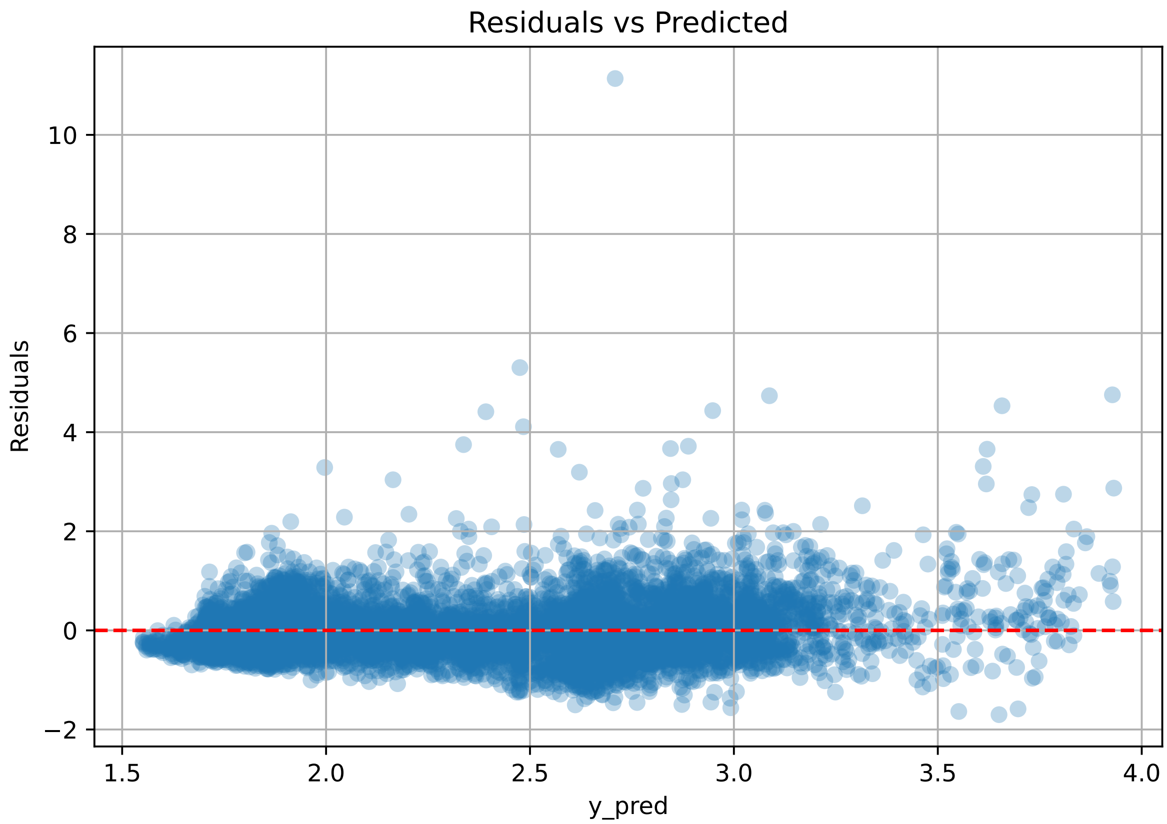
Rank 10: {'reg__iterations': 900, 'reg__depth': 4, 'reg__learning_rate': 0.028709114560267365, 'reg__l2_leaf_reg': 9.275623158132397, 'reg__bagging_temperature': 0.15237877638669295, 'reg__random_strength': 2.613467292764253, 'reg__subsample': 0.5449020519399341, 'reg__min_data_in_leaf': 34, 'reg__leaf_estimation_iterations': 1}

2.5.2 Scatter



Predicted vs True
perfect prediction
y_pred
y_true
14
12
10
8
6
4
2
2
4
6
8
10
12
14

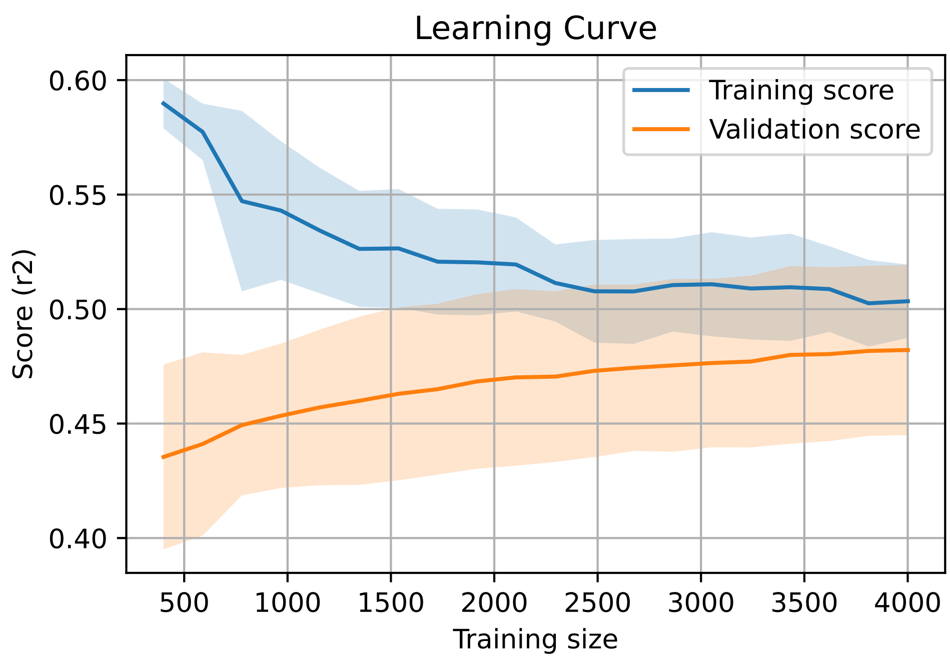
2.5.3 Scatter residuals



2.5.4 Learning curve

Len(group) : 8168, **Len(training)** : 5718,

Len(training_real) : 4003, **Len(test_of_training)** : 1715, **Len(test)** : 2450



2.6 Sumup-table

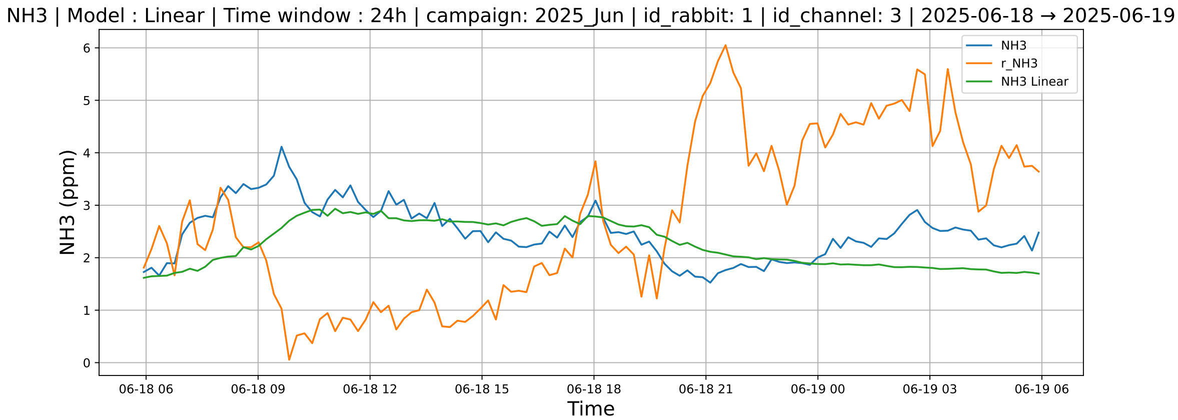
model	mean_train_cv	mean_test_cv	std_test_cv	test_cv	test	diff	r_test	r_train	pen_score
Linear	0.36	0.365	0.037	0.345	0.345	-0.02	0.946	0.986	nan
LGBM	0.464	0.444	0.037	0.454	0.48	0.036	1.081	1.046	0.4439
CatBoost	0.506	0.483	0.037	0.501	0.506	0.023	1.048	1.047	0.483

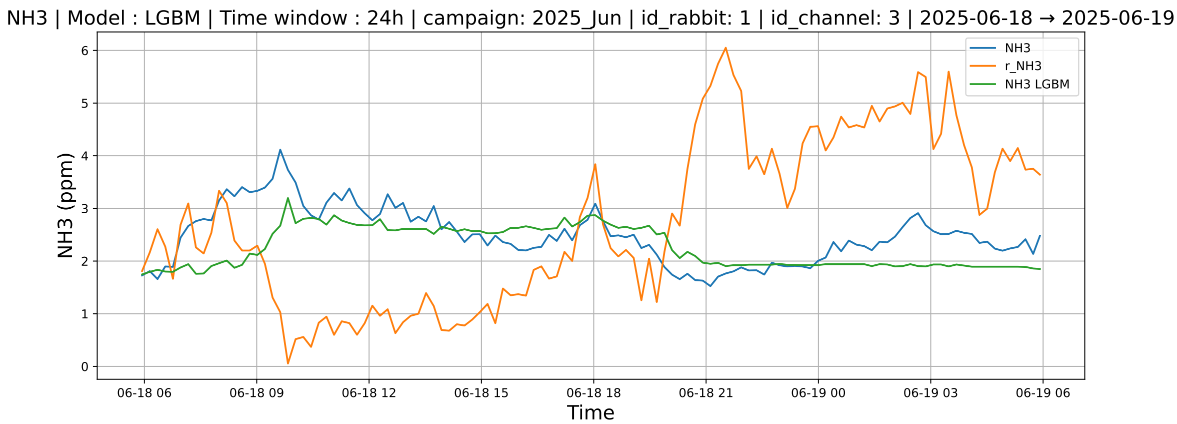
3 Plot

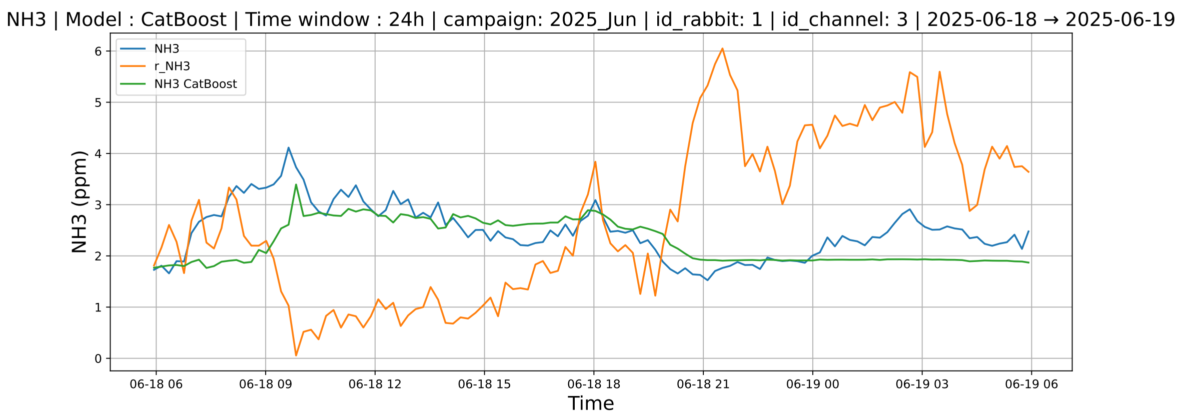
3.1 Standalone plots

3.1.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

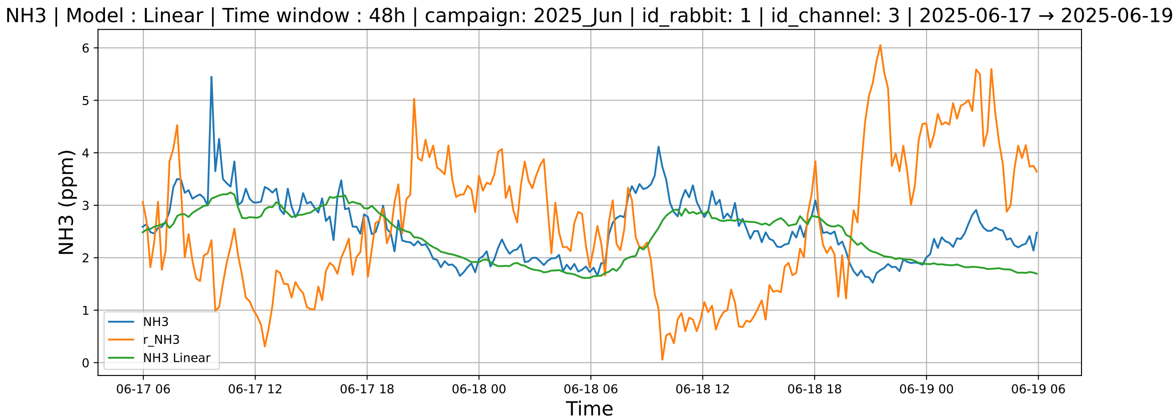
3.1.1.1 Time window : 24

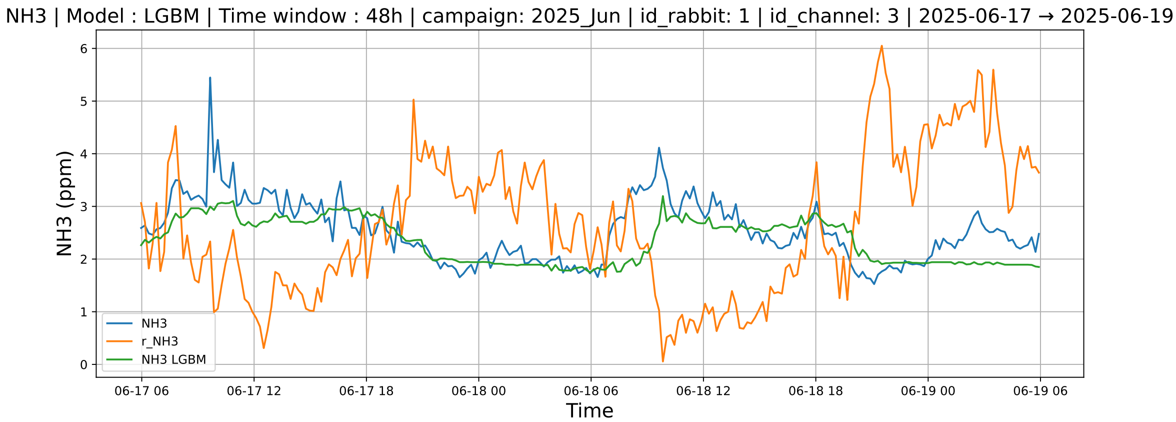


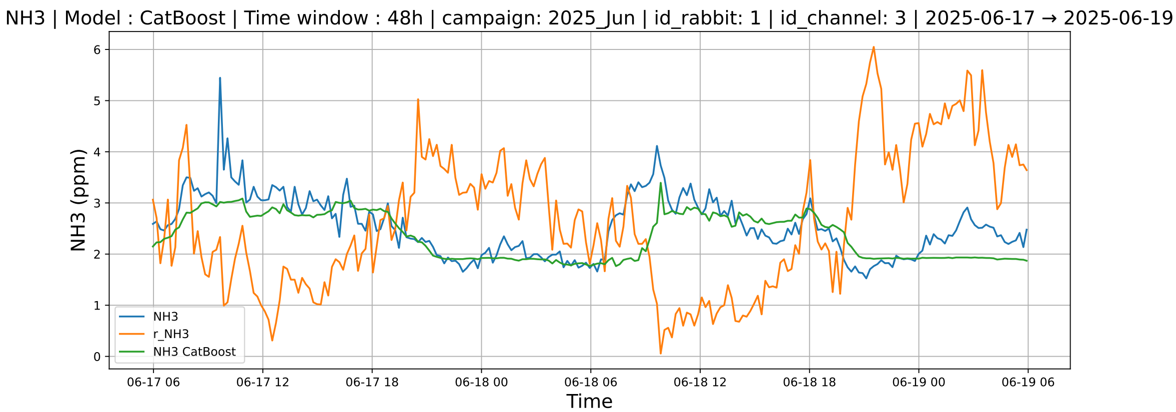




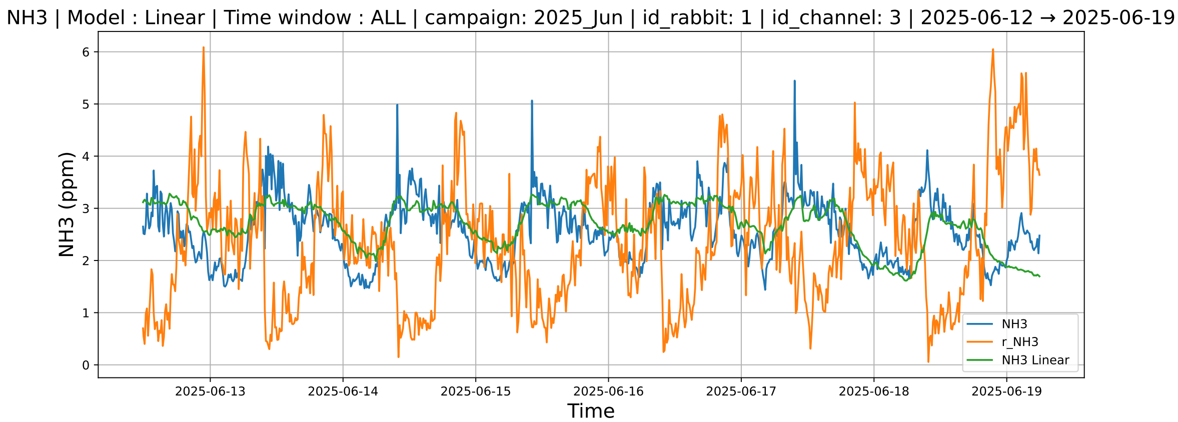
3.1.1.2 Time window : 48

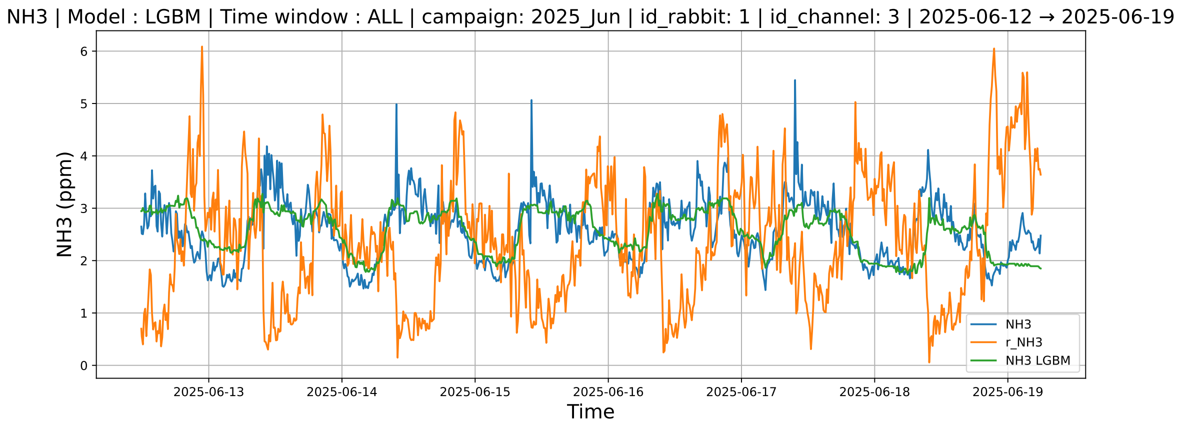


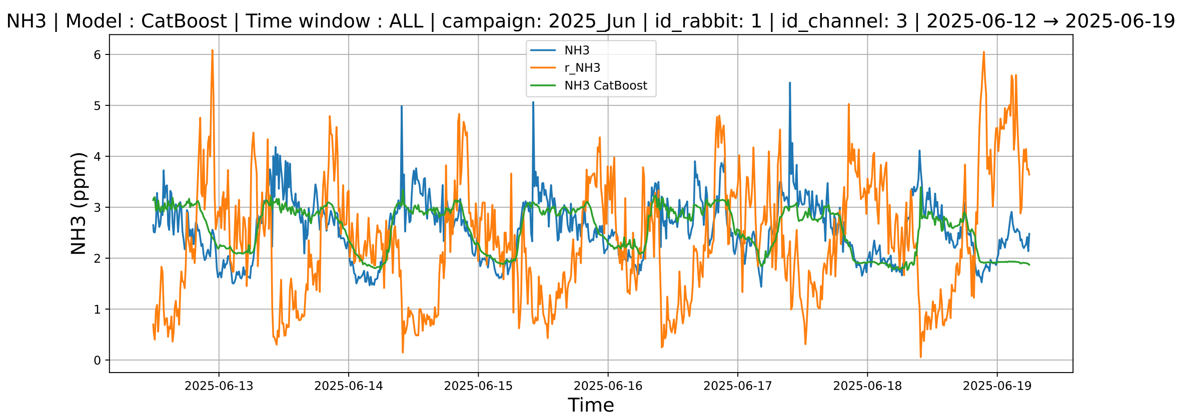




3.1.1.3 Time window : ALL

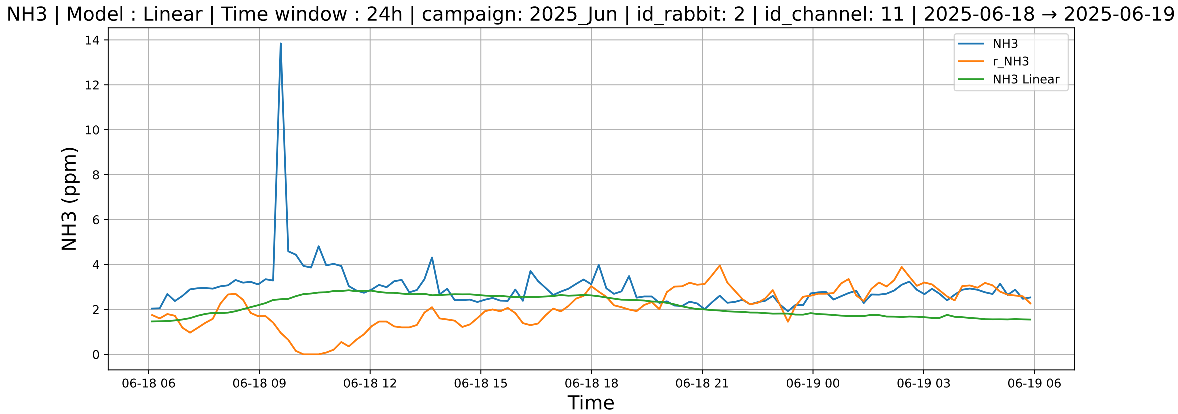


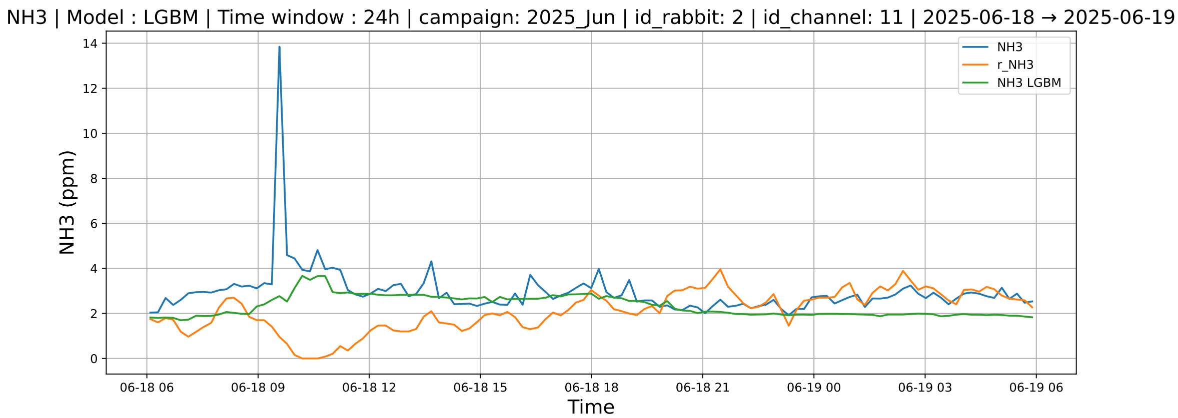


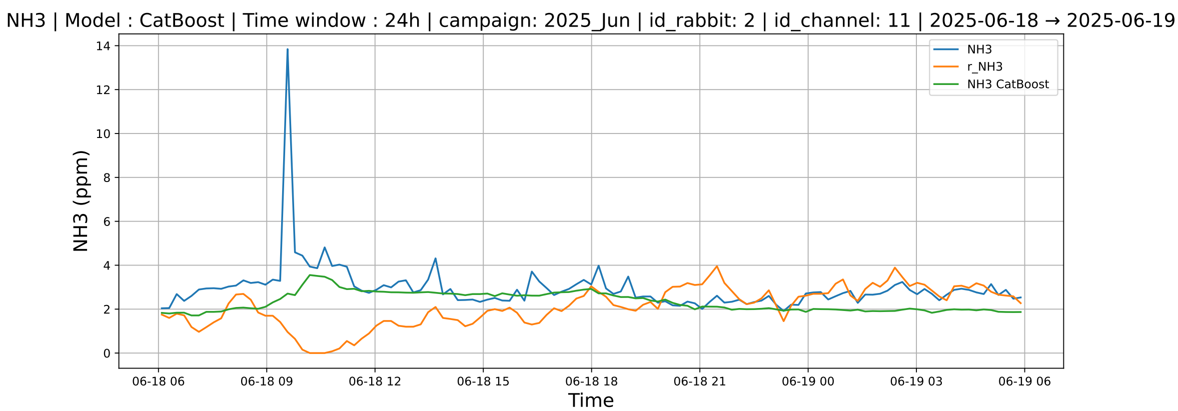


3.1.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

3.1.2.1 Time window : 24

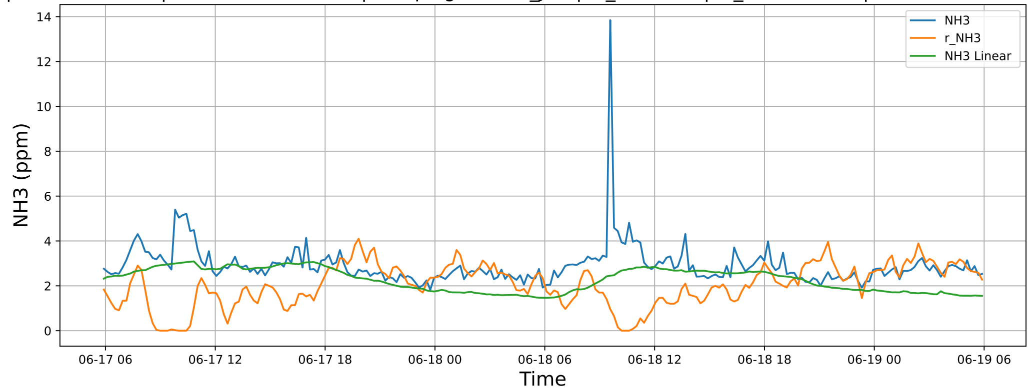




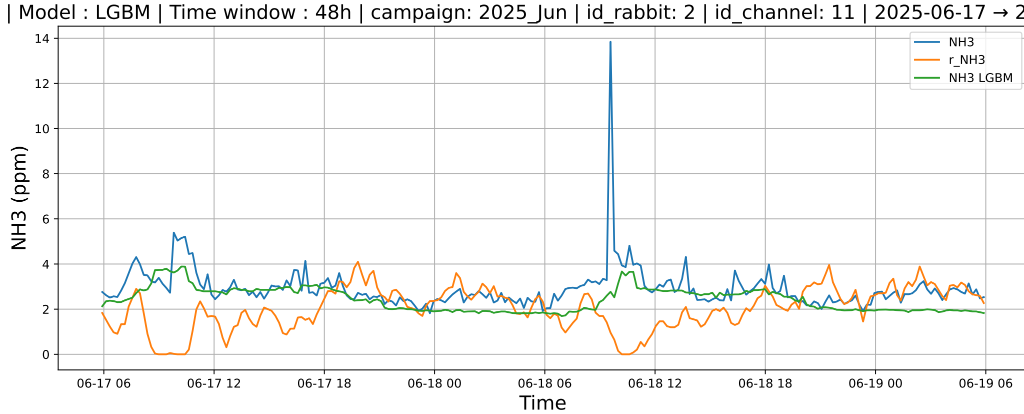


3.1.2.2 Time window : 48

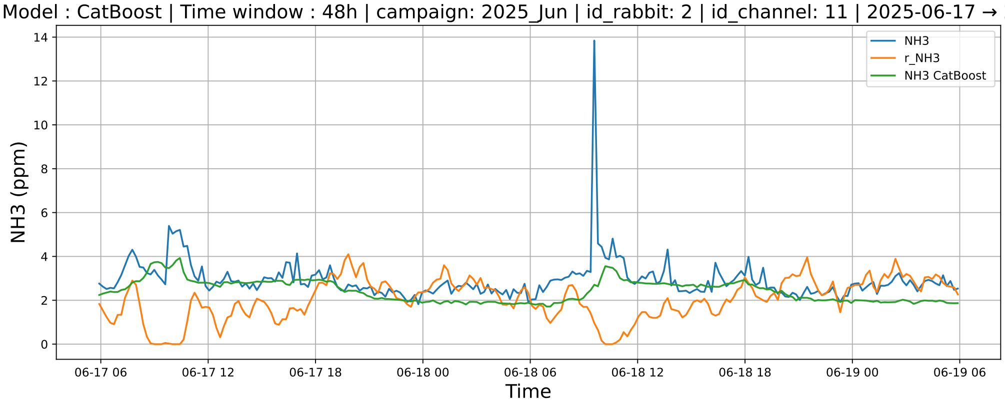
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

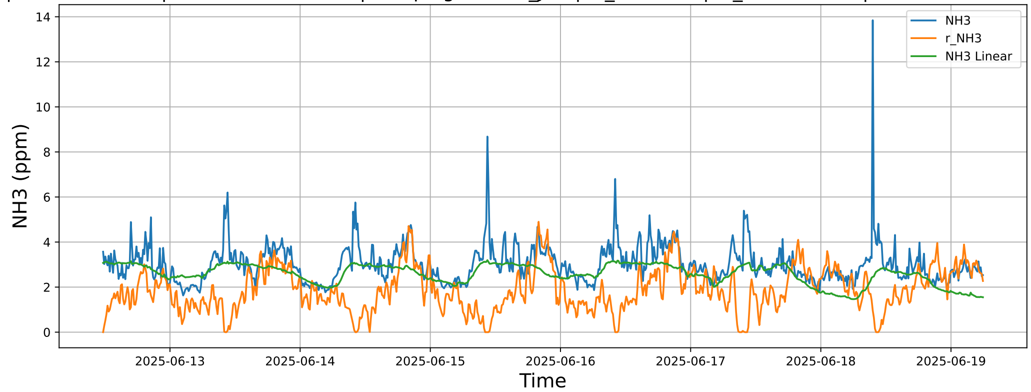


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-17 → 2025-06-19

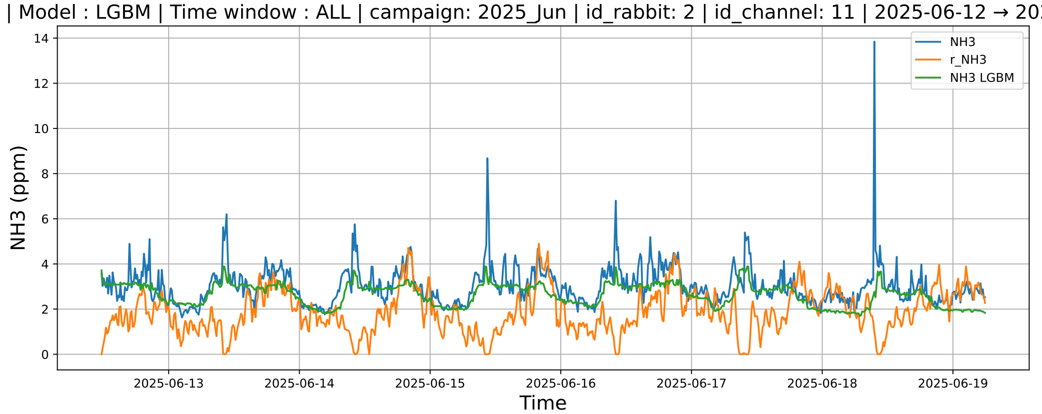


3.1.2.3 Time window : ALL

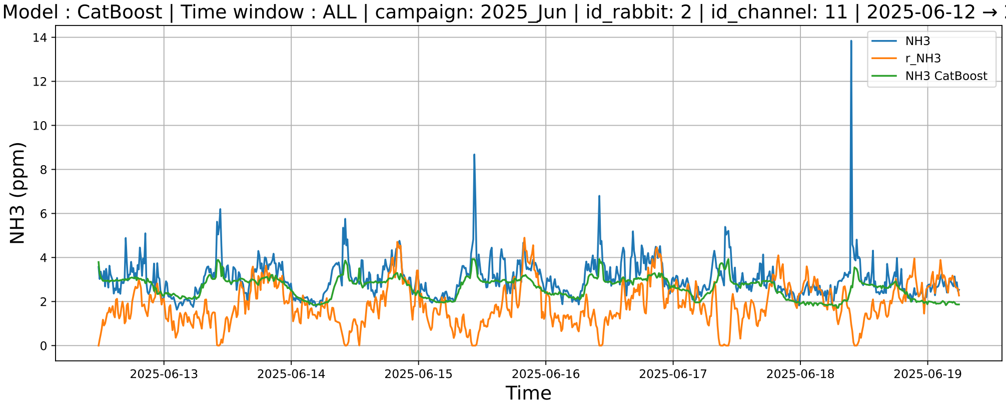
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

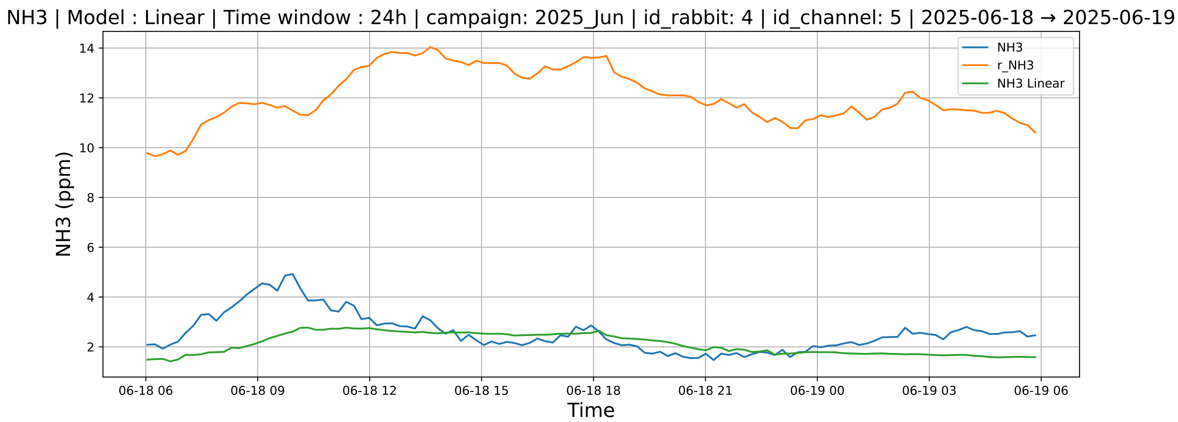


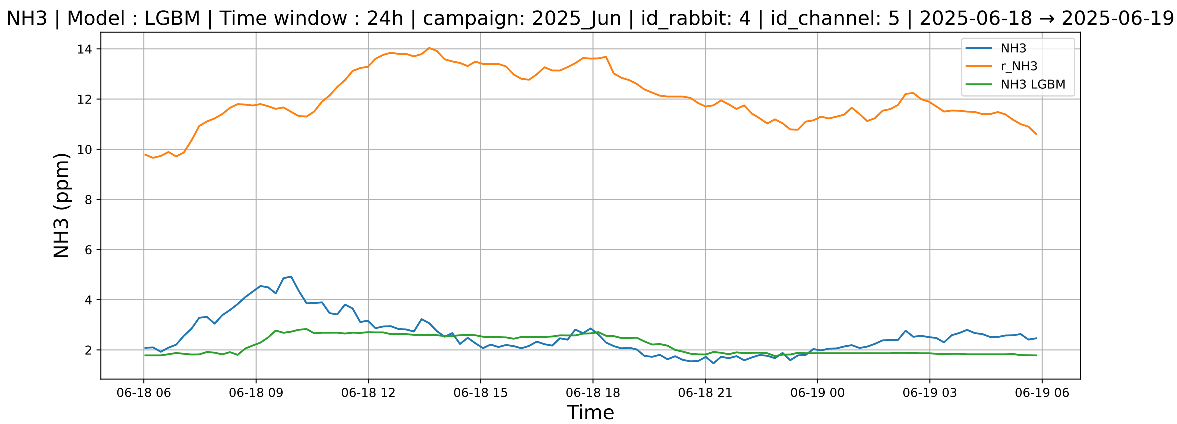
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 2 | id_channel: 11 | 2025-06-12 → 2025-06-19

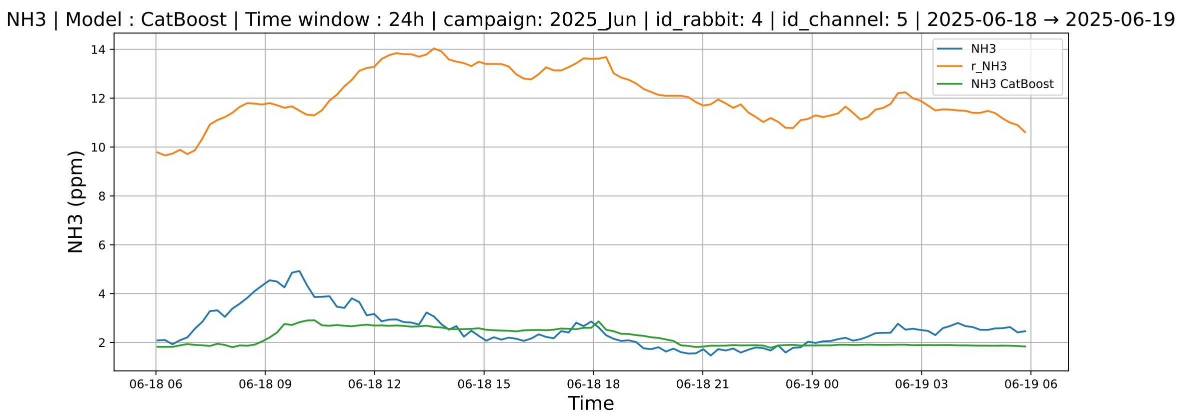


3.1.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

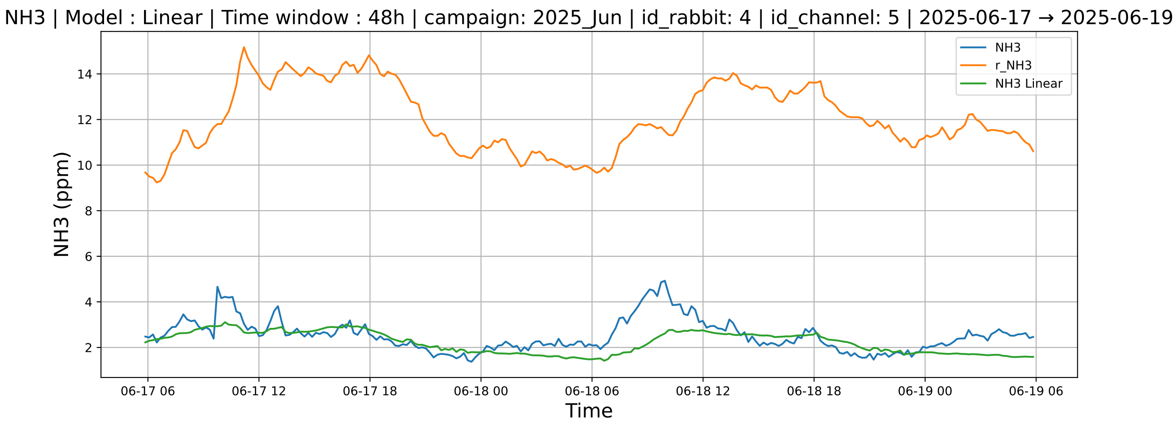
3.1.3.1 Time window : 24

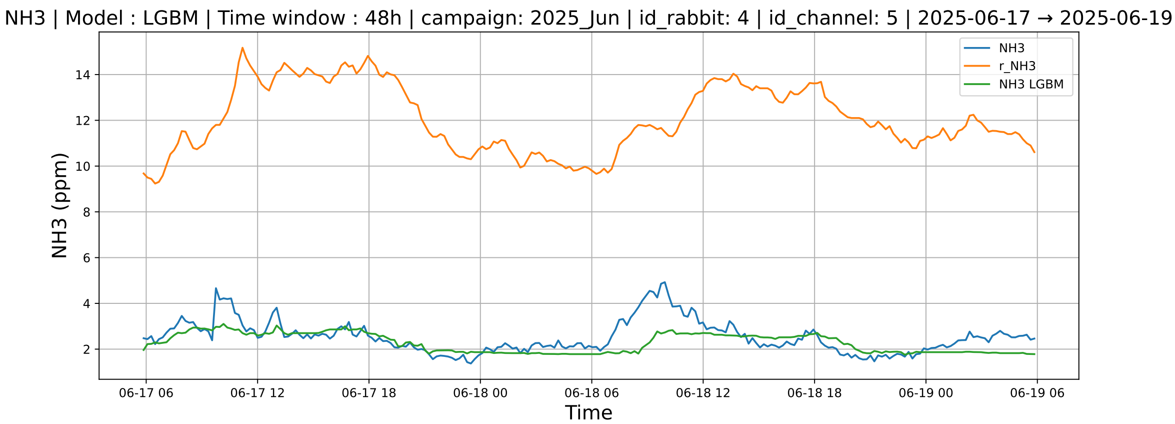


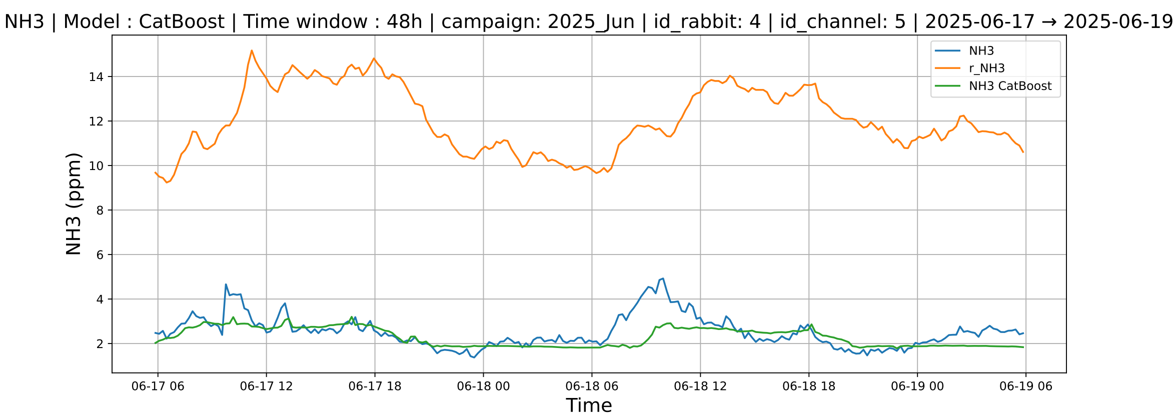




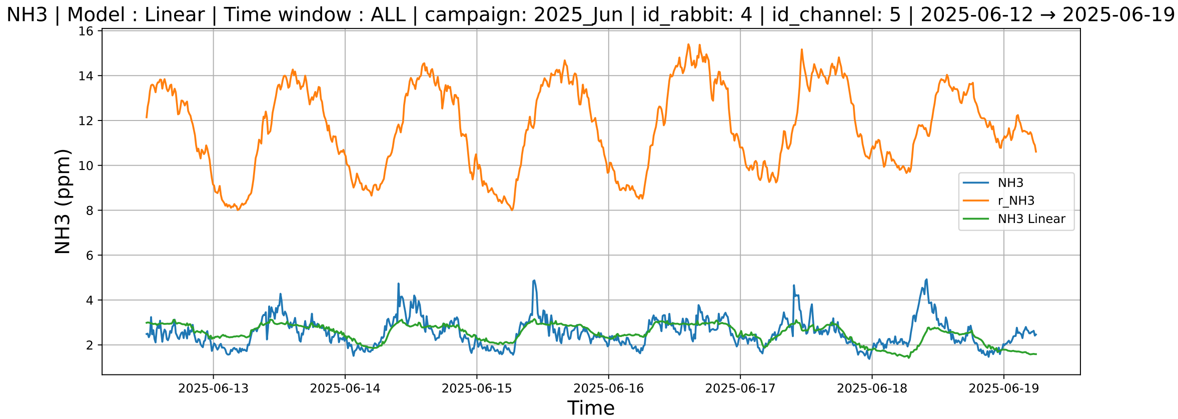
3.1.3.2 Time window : 48

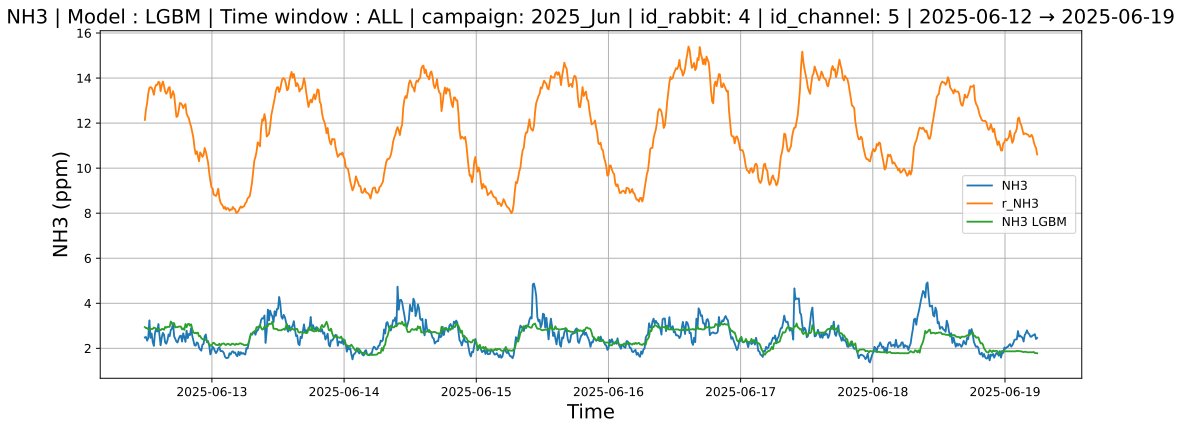


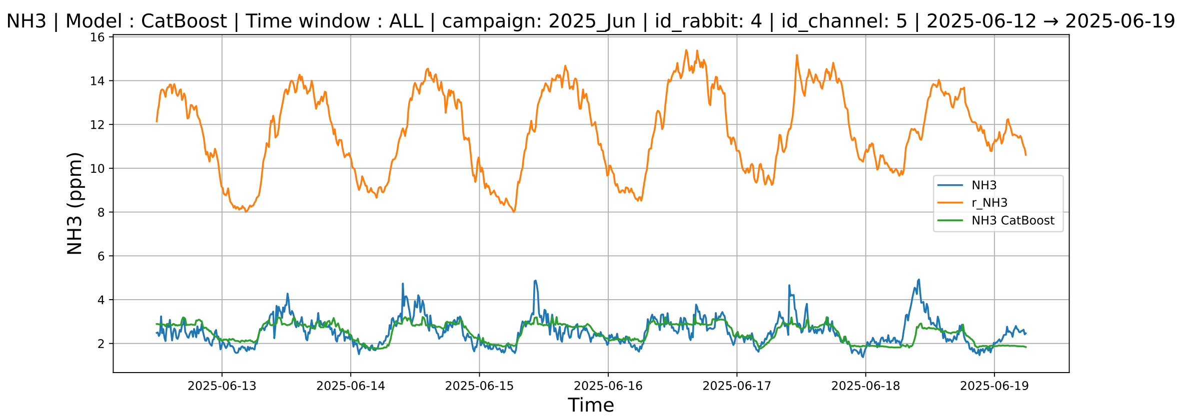




3.1.3.3 Time window : ALL



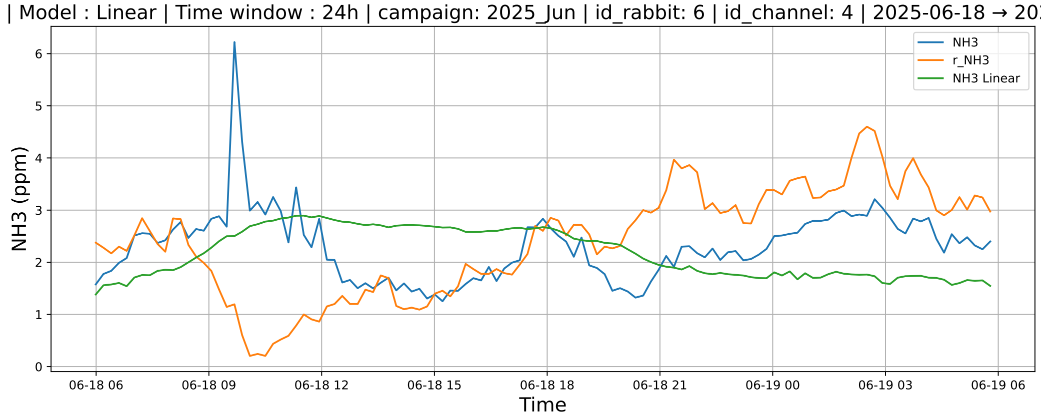




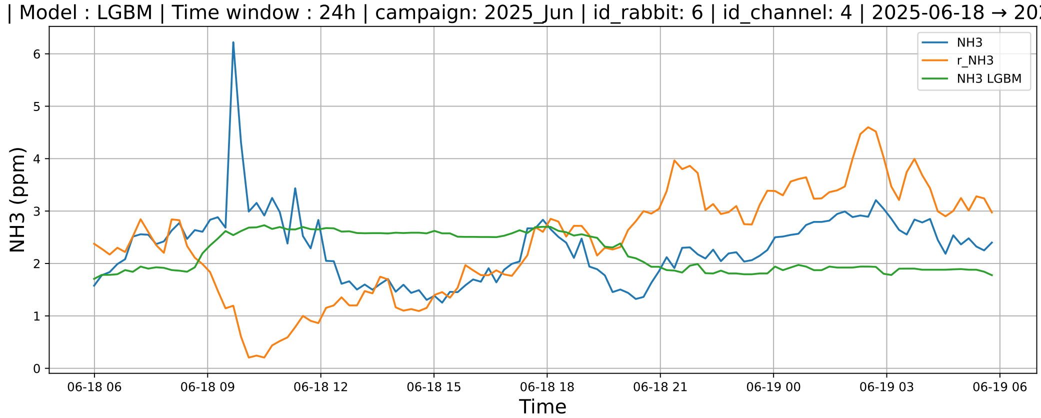
3.1.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

3.1.4.1 Time window : 24

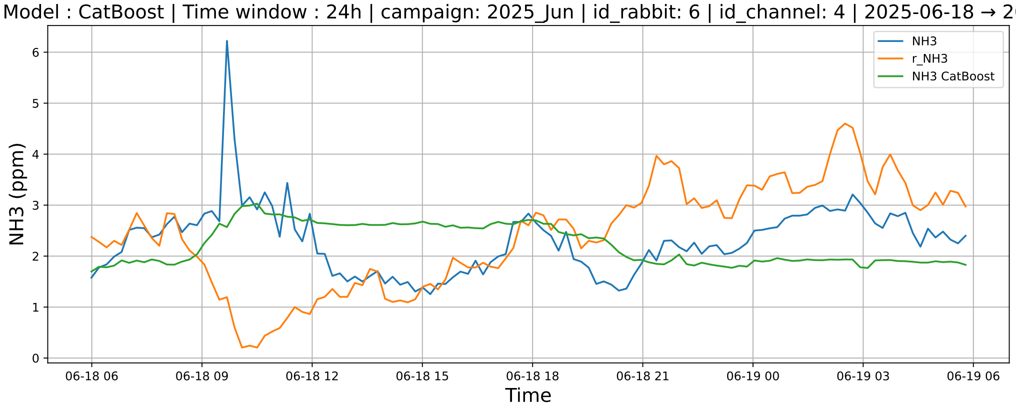
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19



NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

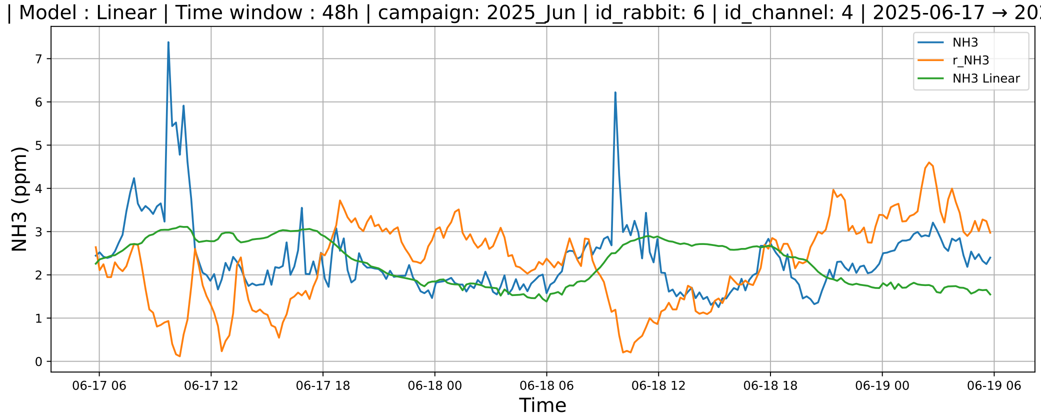


NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-18 → 2025-06-19

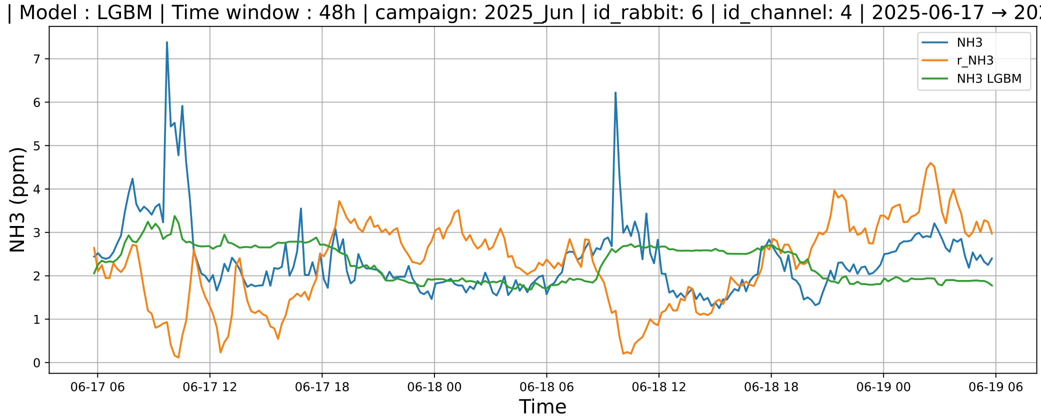


3.1.4.2 Time window : 48

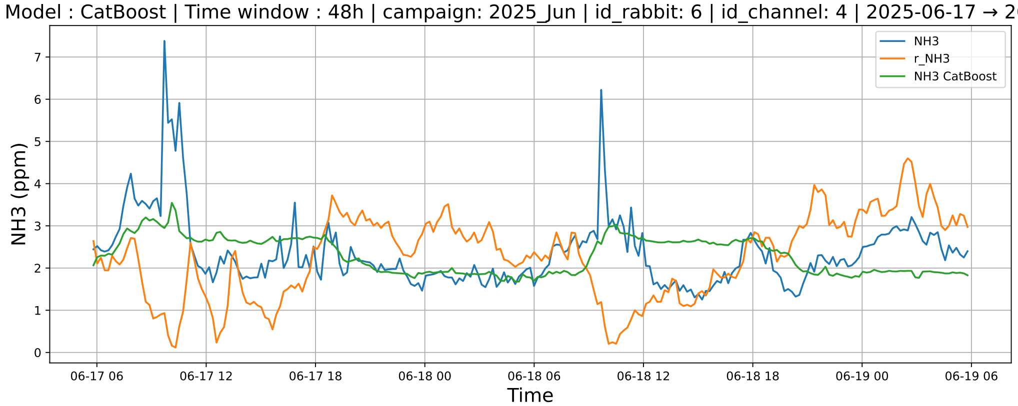
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19



NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

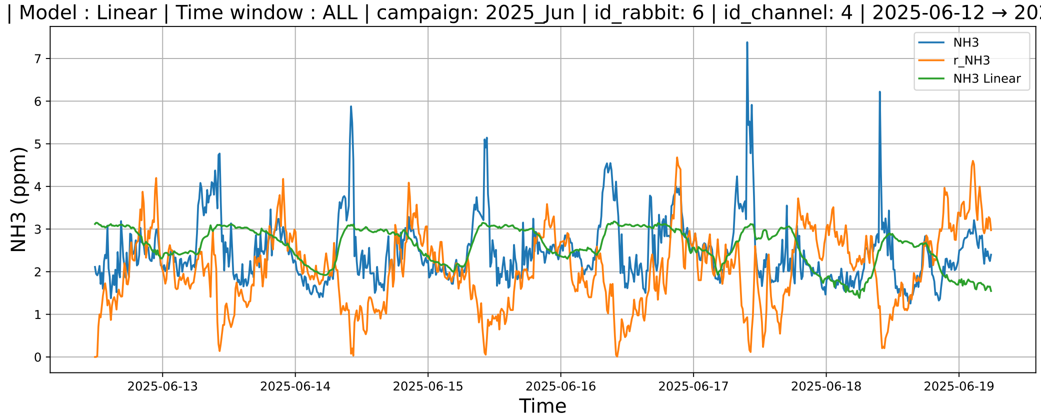


NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-17 → 2025-06-19

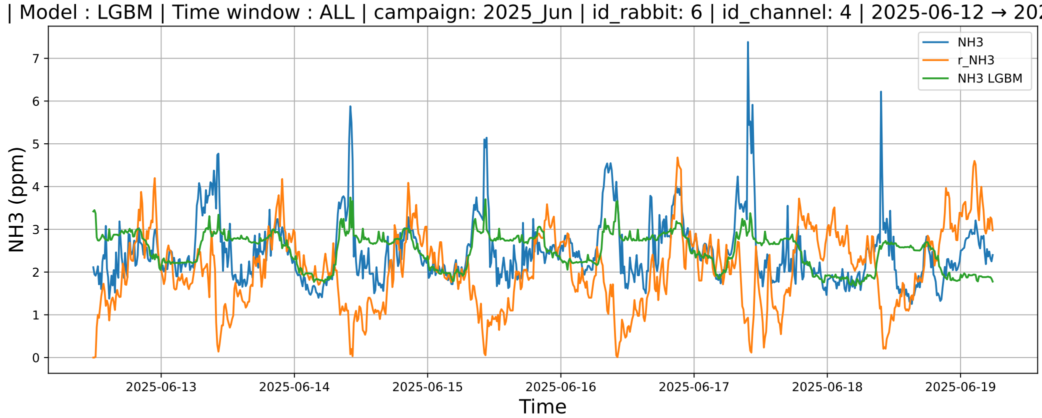


3.1.4.3 Time window : ALL

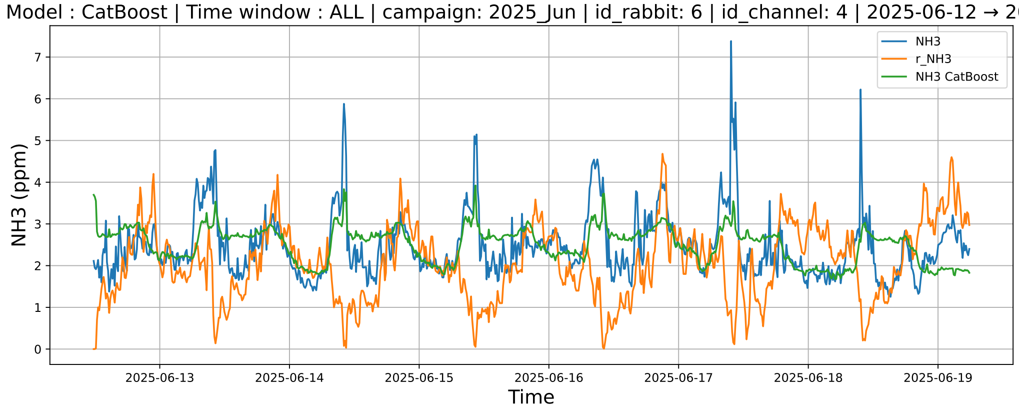
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



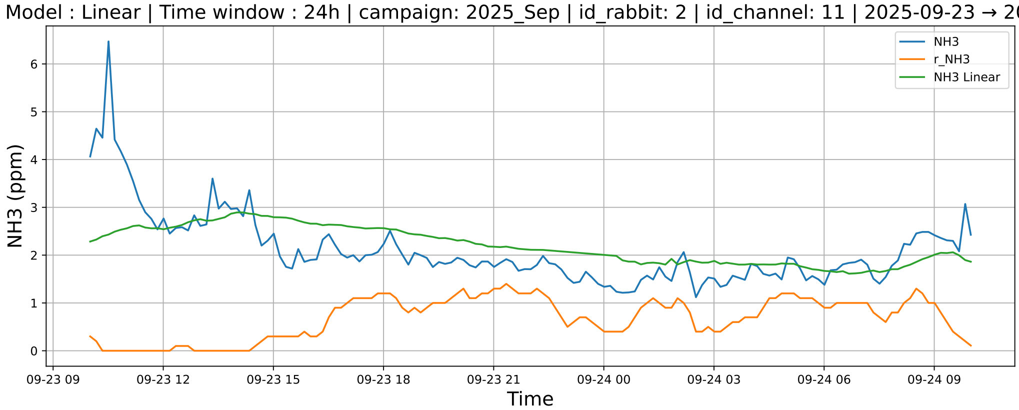
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Jun | id_rabbit: 6 | id_channel: 4 | 2025-06-12 → 2025-06-19



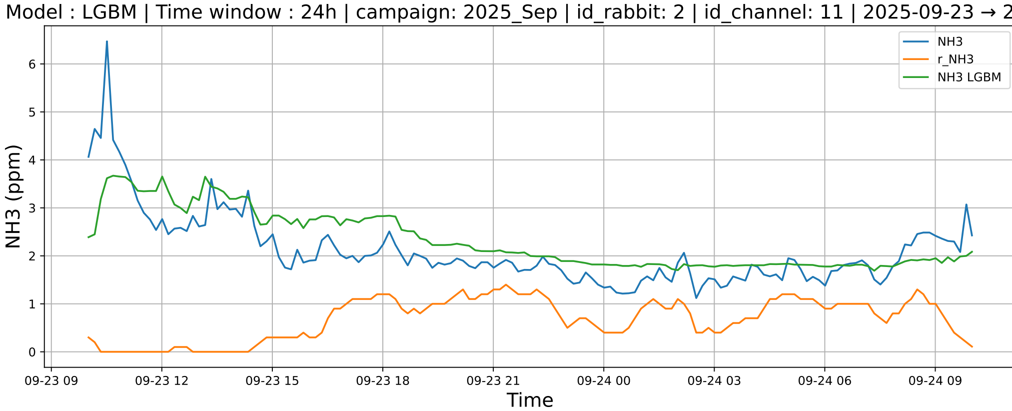
3.1.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

3.1.5.1 Time window : 24

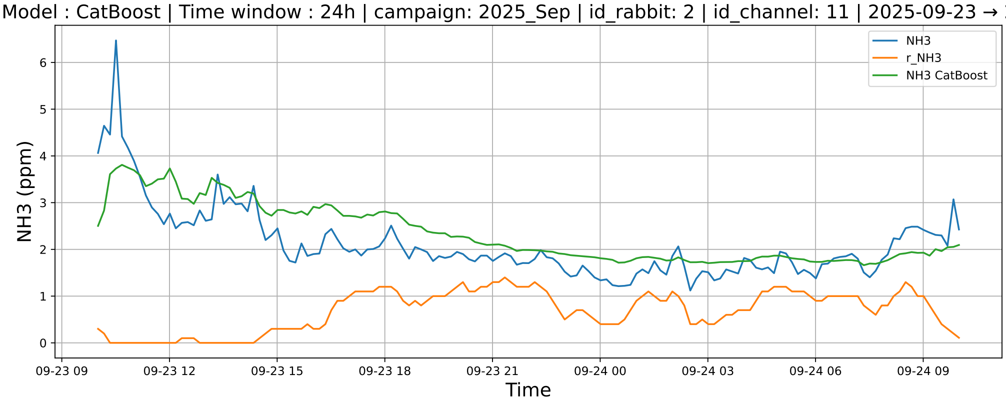
NH3 | Model : Linear | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



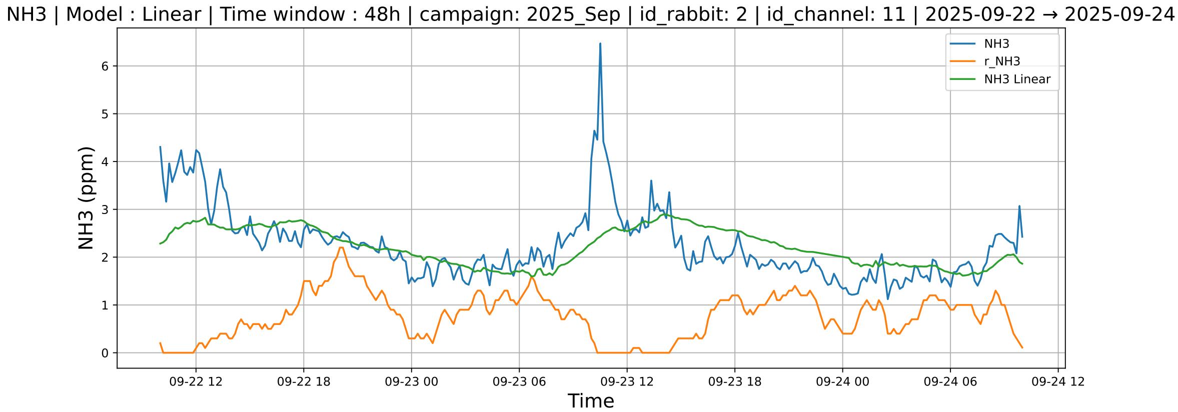
NH3 | Model : LGBM | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24

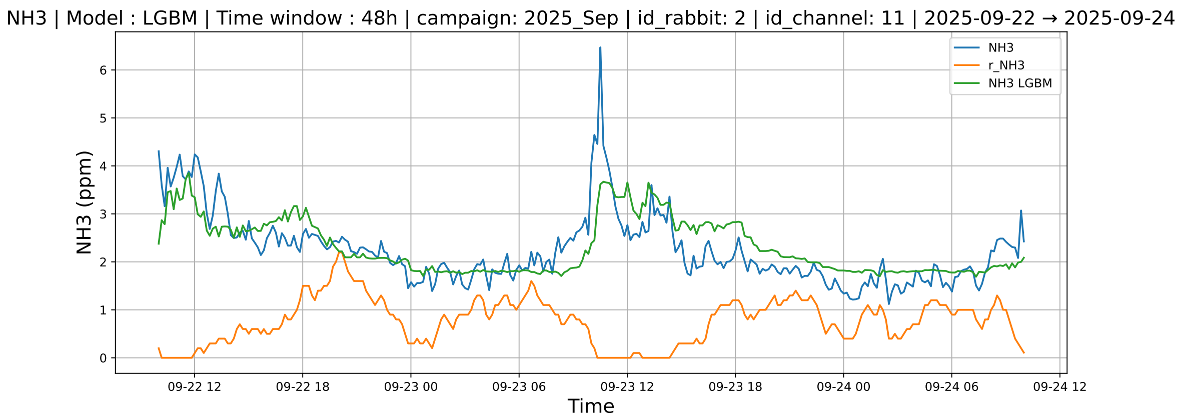


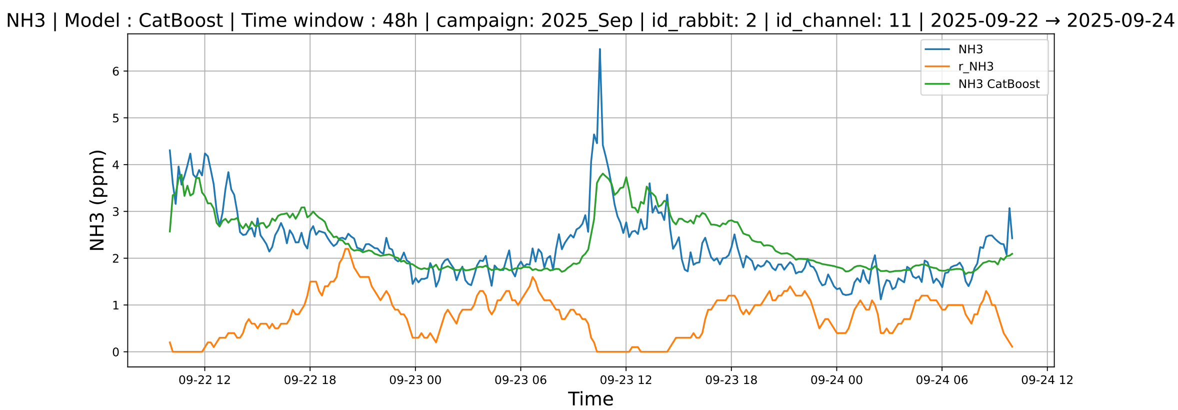
NH3 | Model : CatBoost | Time window : 24h | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-23 → 2025-09-24



3.1.5.2 Time window : 48

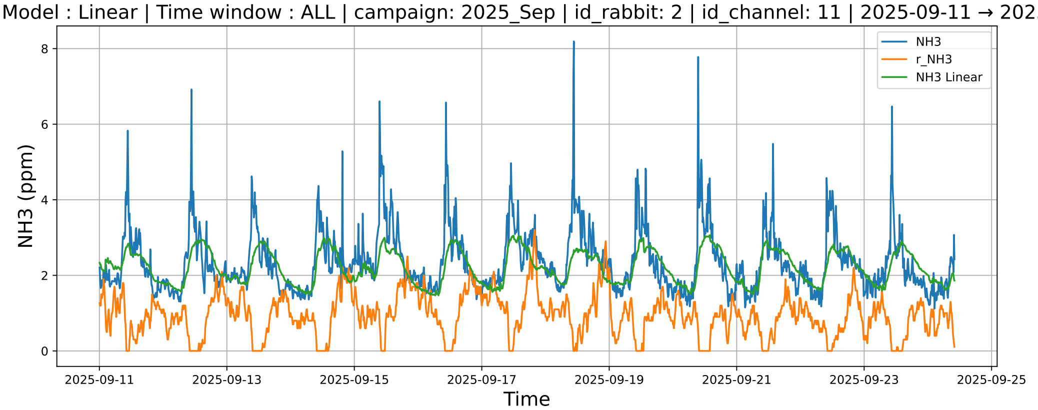




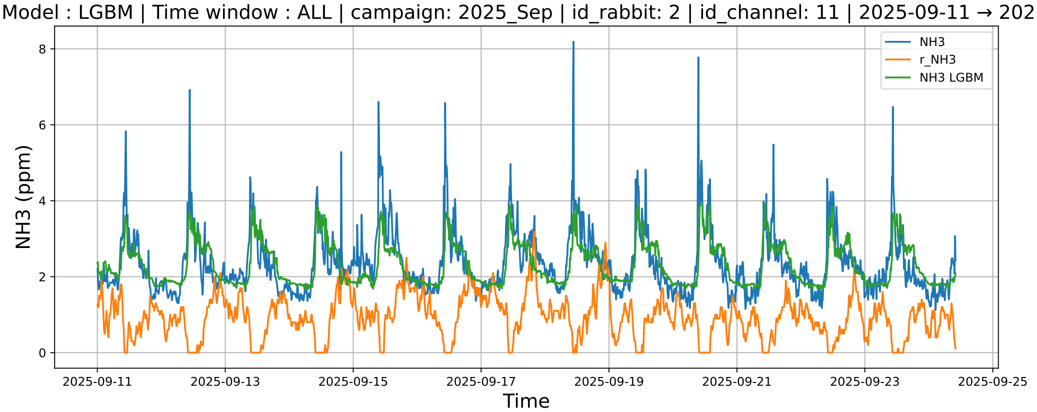


3.1.5.3 Time window : ALL

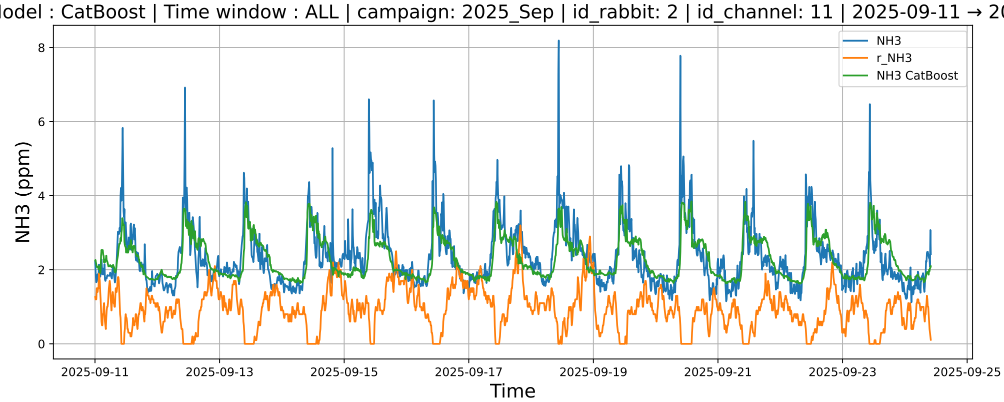
NH3 | Model : Linear | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24



NH3 | Model : LGBM | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

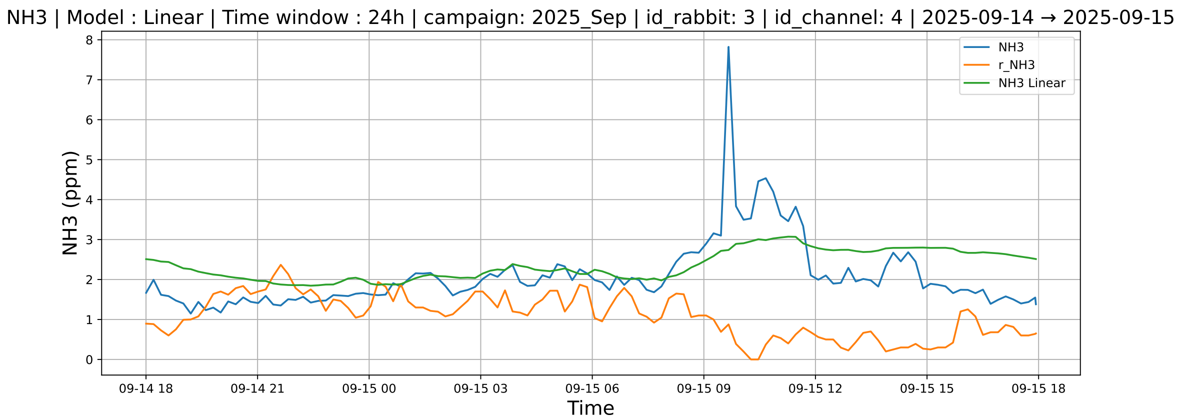


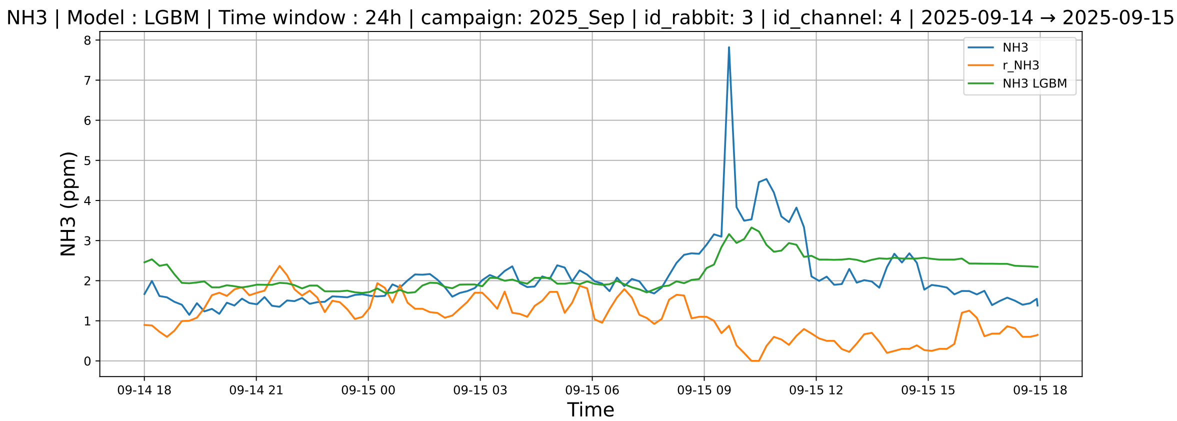
NH3 | Model : CatBoost | Time window : ALL | campaign: 2025_Sep | id_rabbit: 2 | id_channel: 11 | 2025-09-11 → 2025-09-24

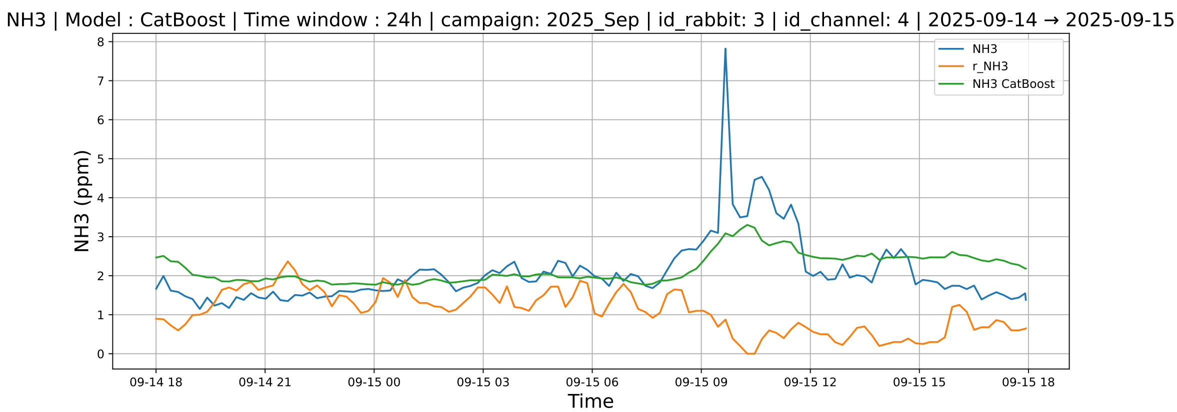


3.1.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

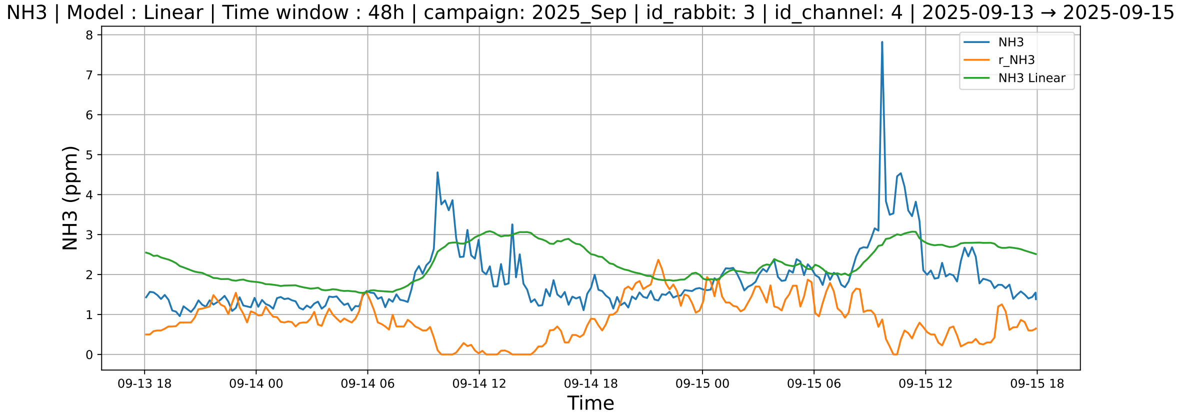
3.1.6.1 Time window : 24

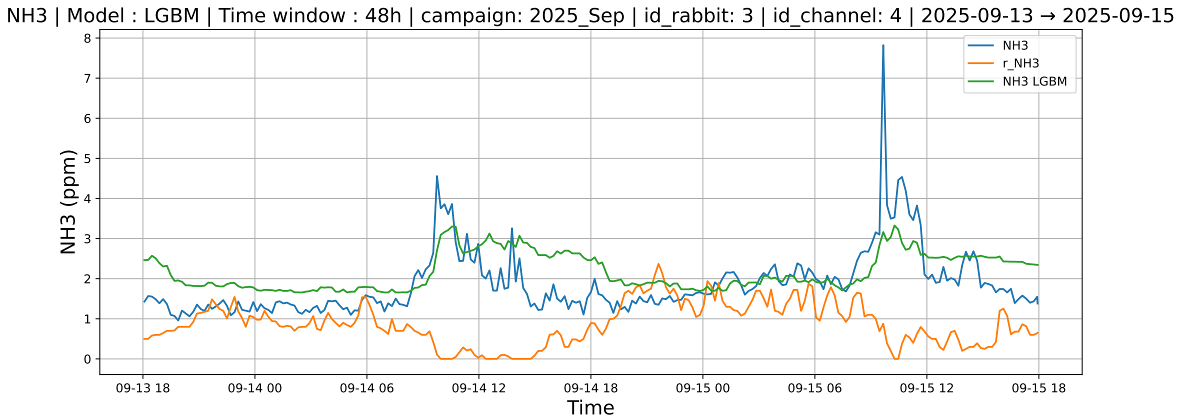


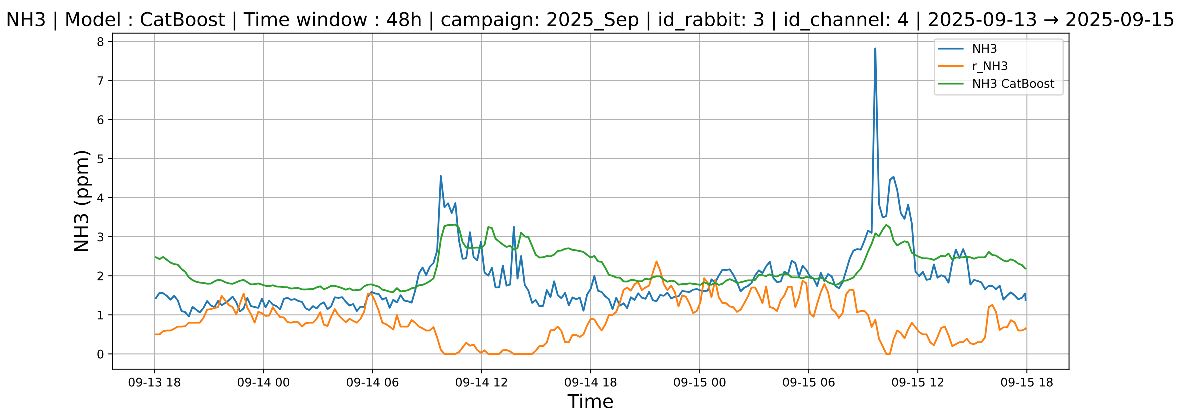




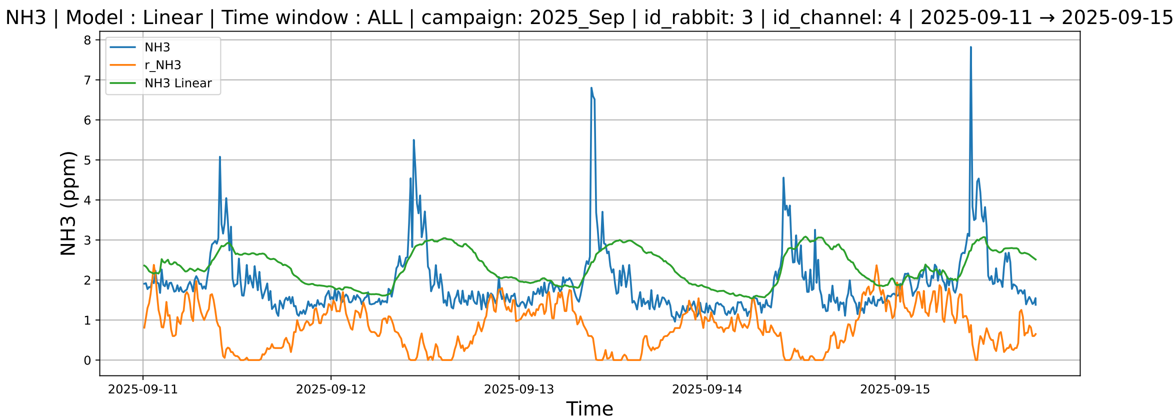
3.1.6.2 Time window : 48

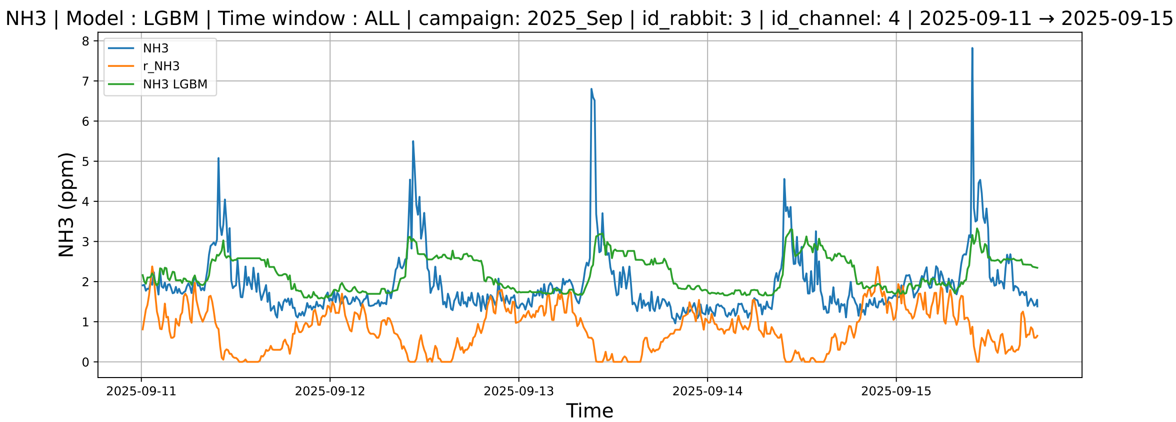


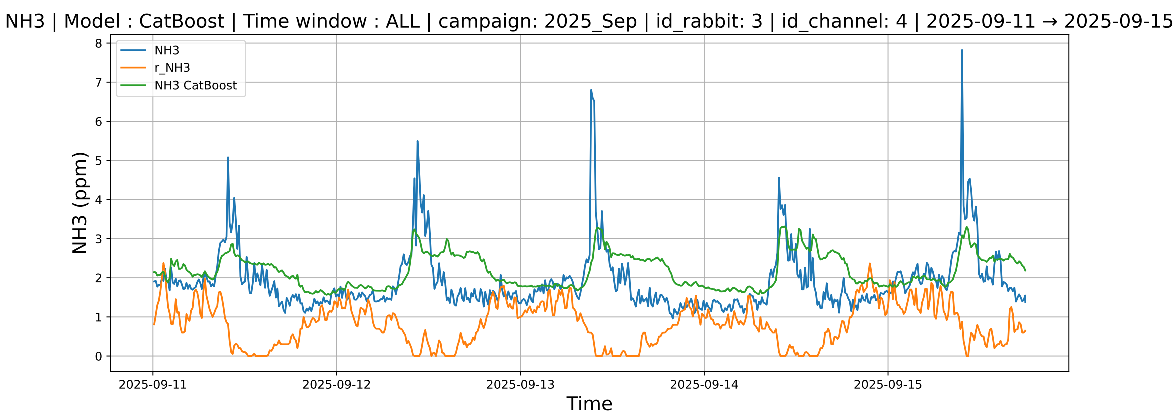




3.1.6.3 Time window : ALL

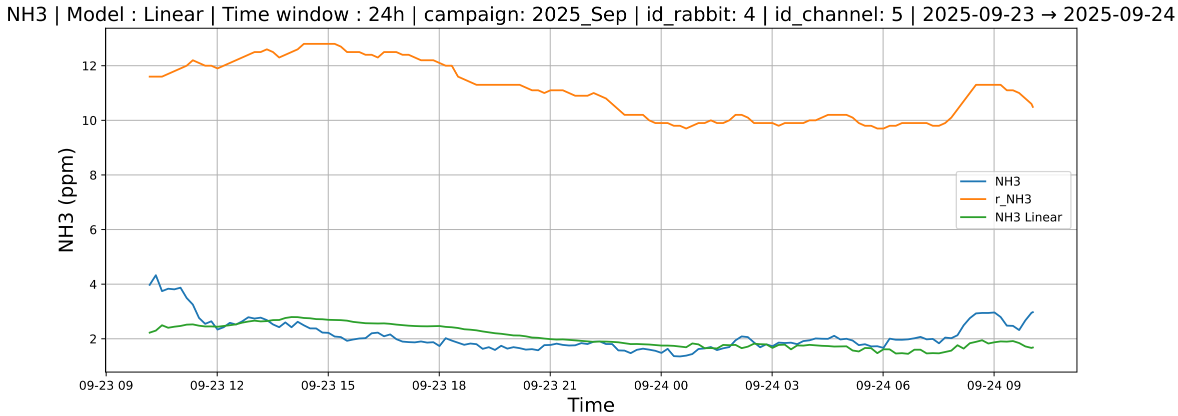


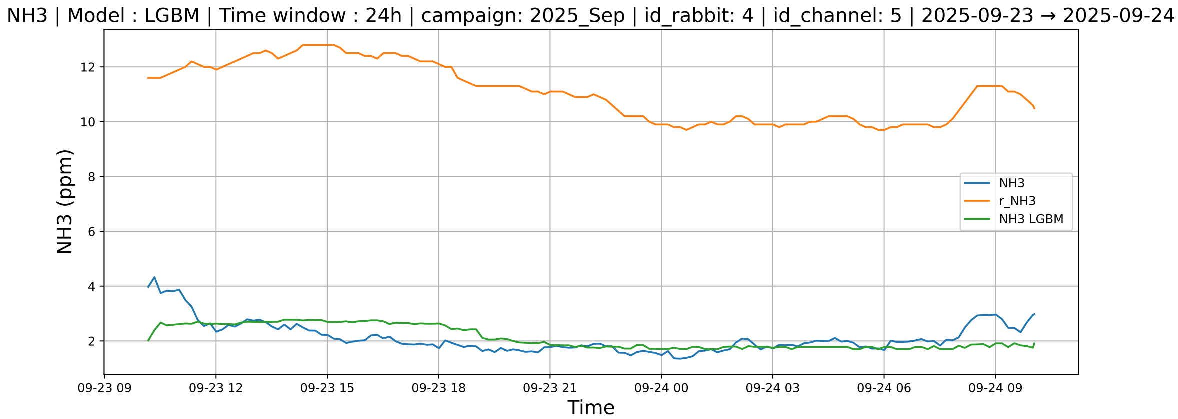


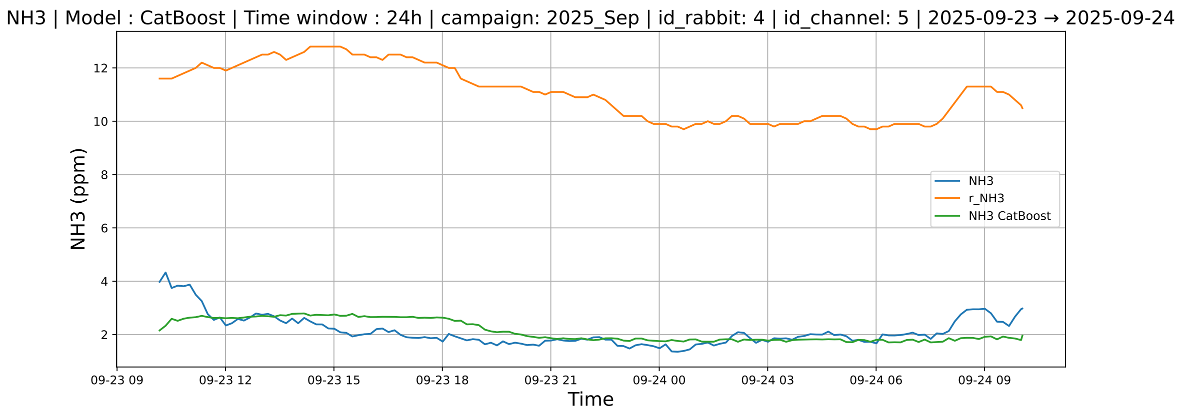


3.1.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

3.1.7.1 Time window : 24

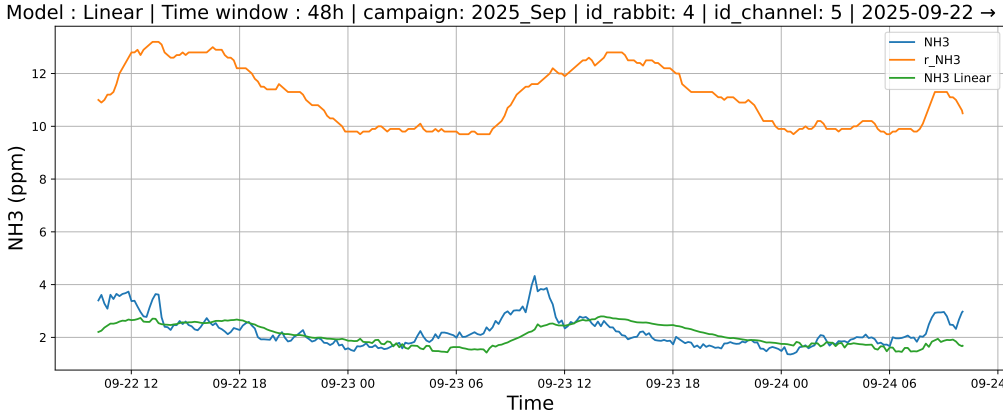




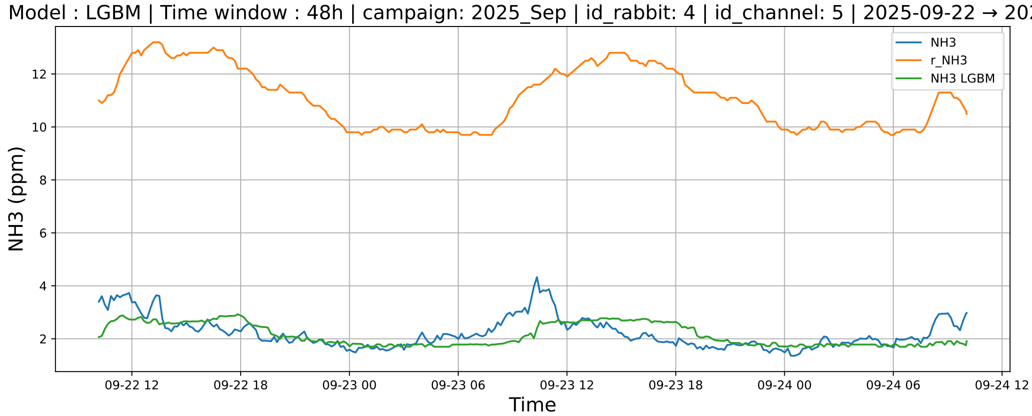


3.1.7.2 Time window : 48

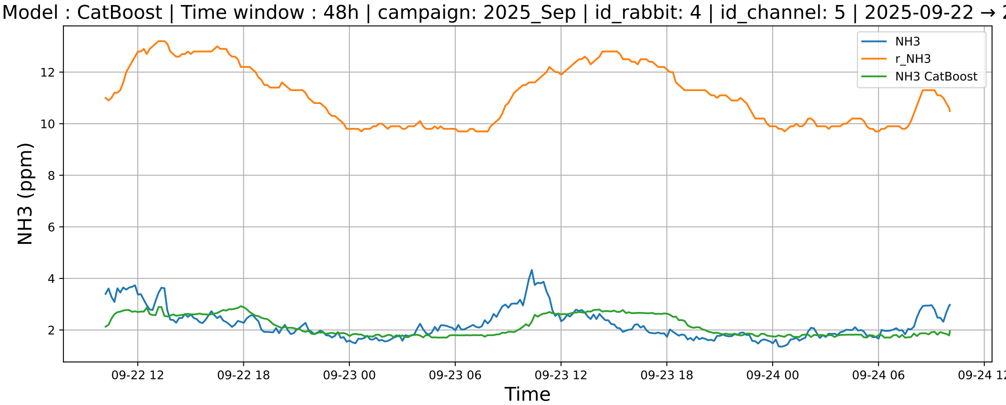
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



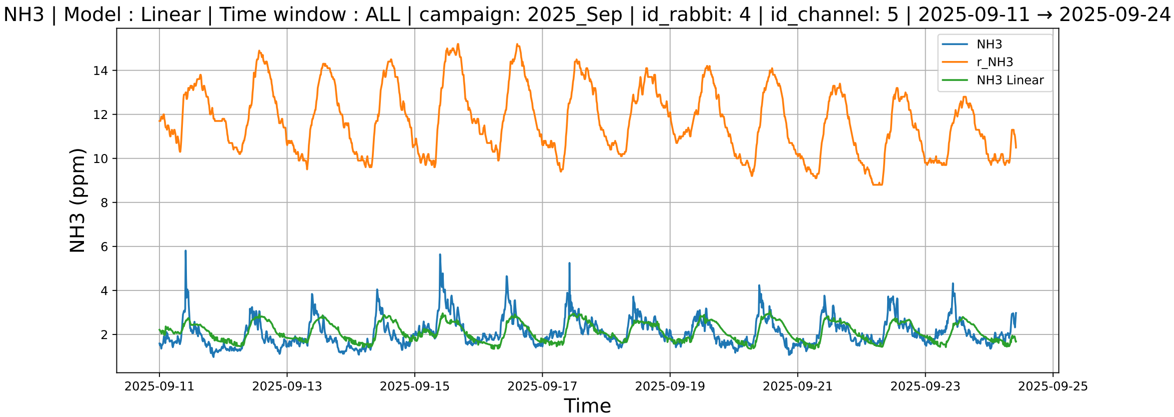
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24

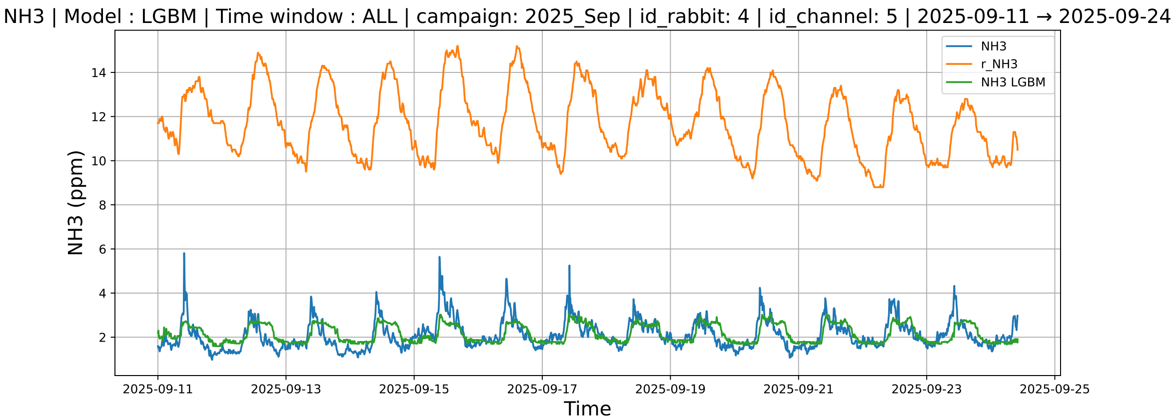


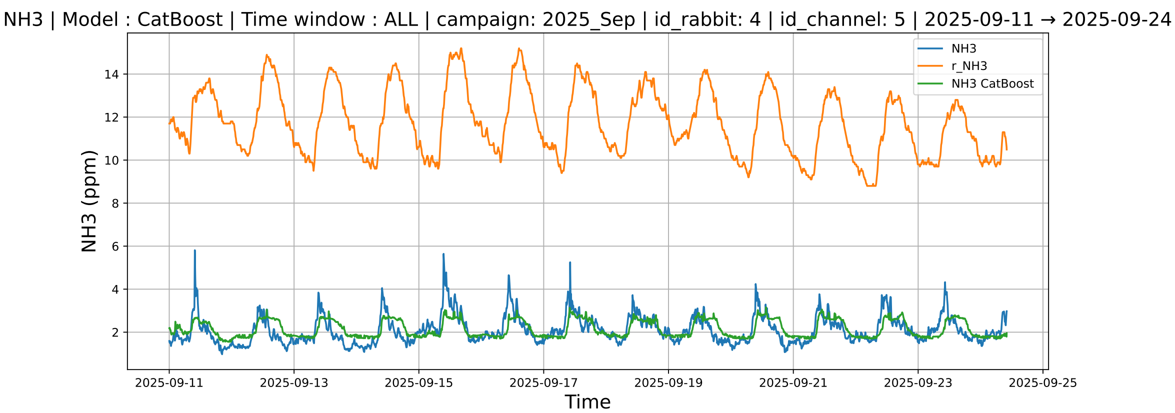
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 4 | id_channel: 5 | 2025-09-22 → 2025-09-24



3.1.7.3 Time window : ALL

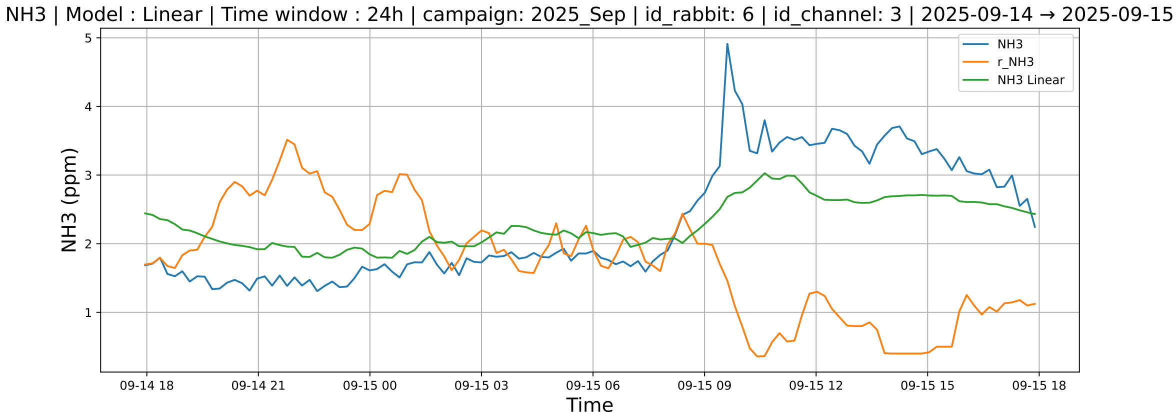


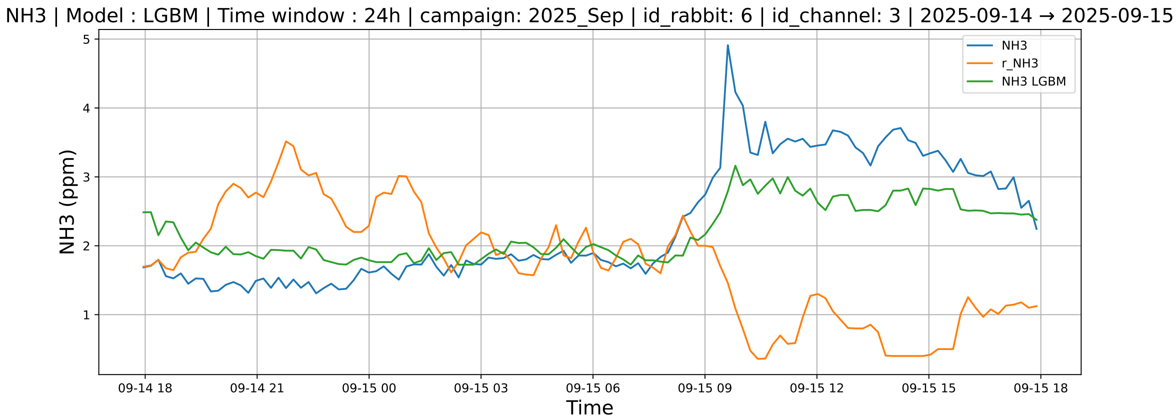


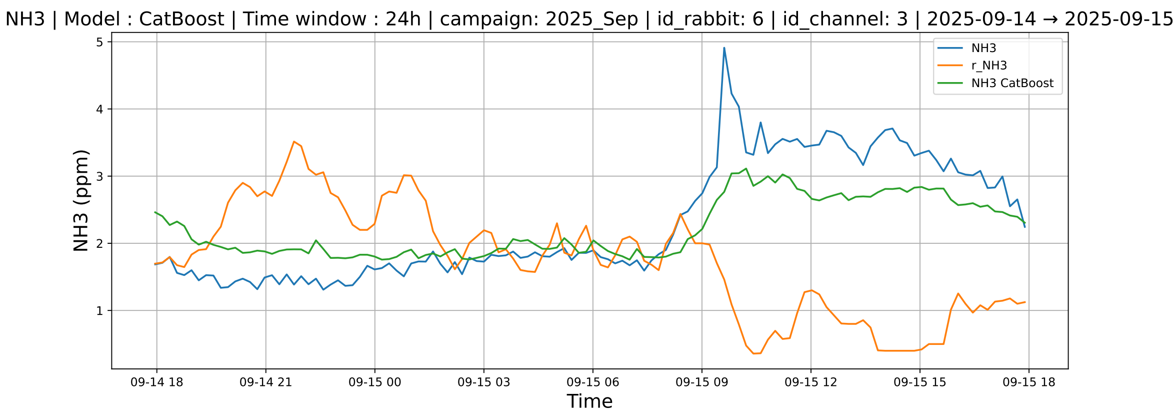


3.1.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.1.8.1 Time window : 24

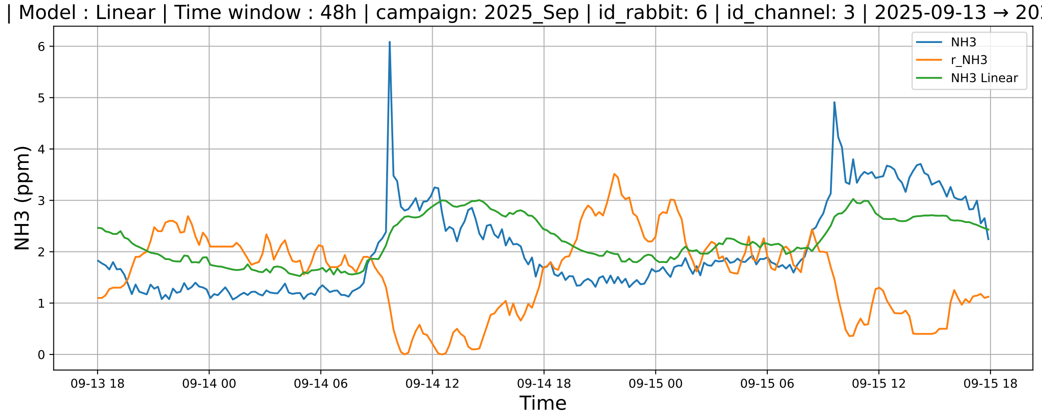




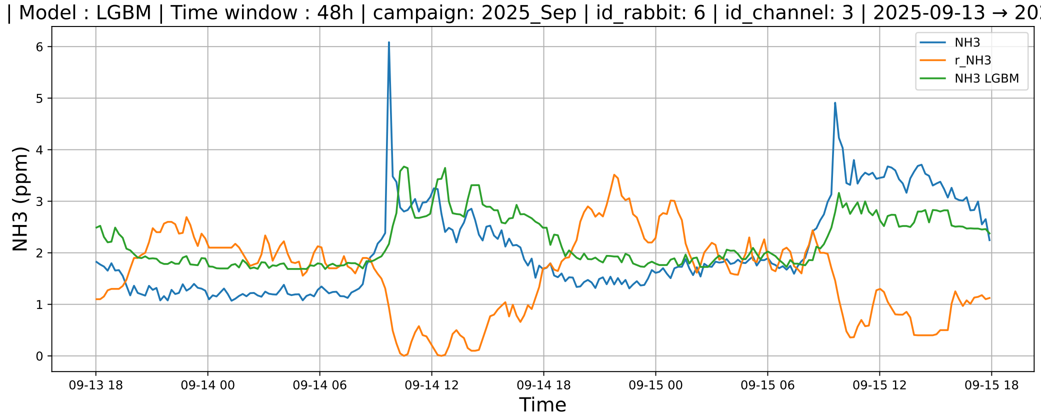


3.1.8.2 Time window : 48

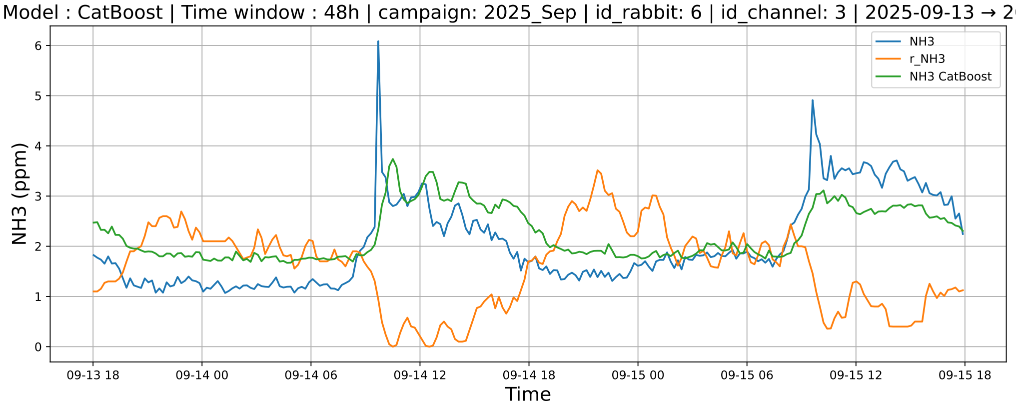
NH3 | Model : Linear | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



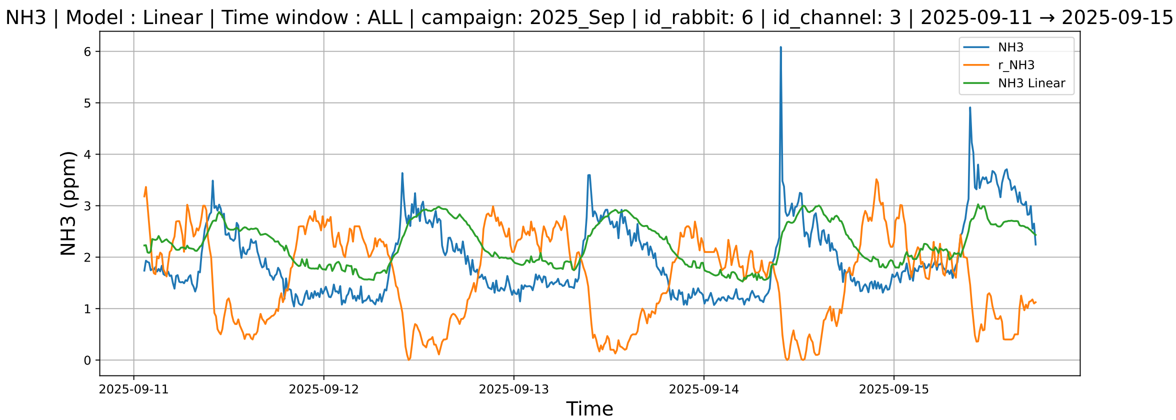
NH3 | Model : LGBM | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15

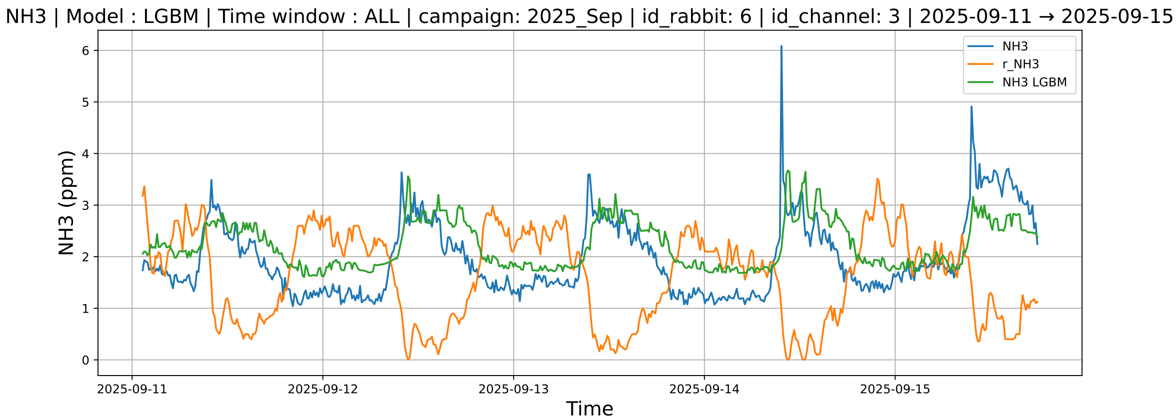


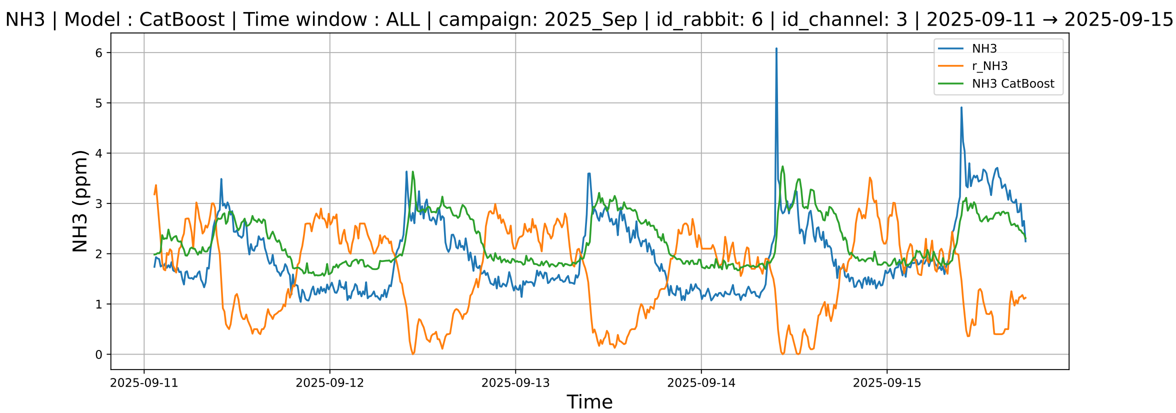
NH3 | Model : CatBoost | Time window : 48h | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-13 → 2025-09-15



3.1.8.3 Time window : ALL





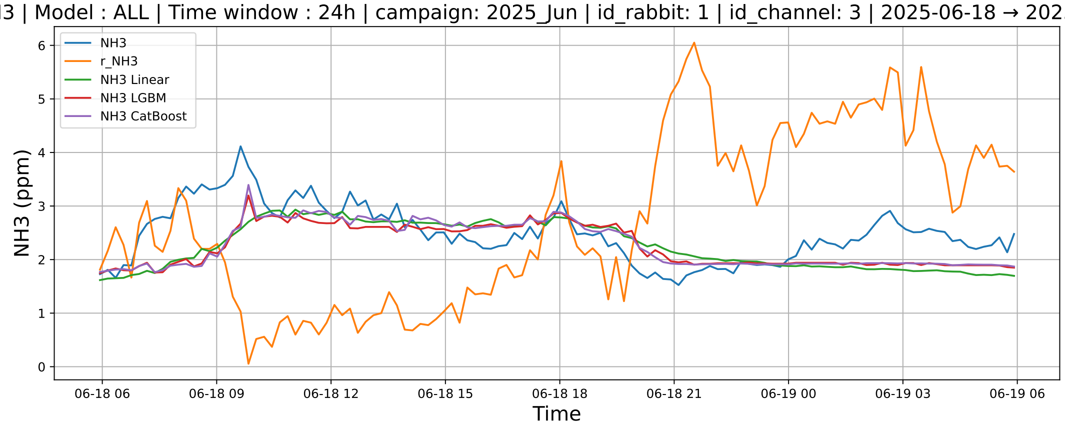


3.2 Compare plots

3.2.1 Group: campaign: 2025_Jun,id_rabbit: 1,id_channel: 3

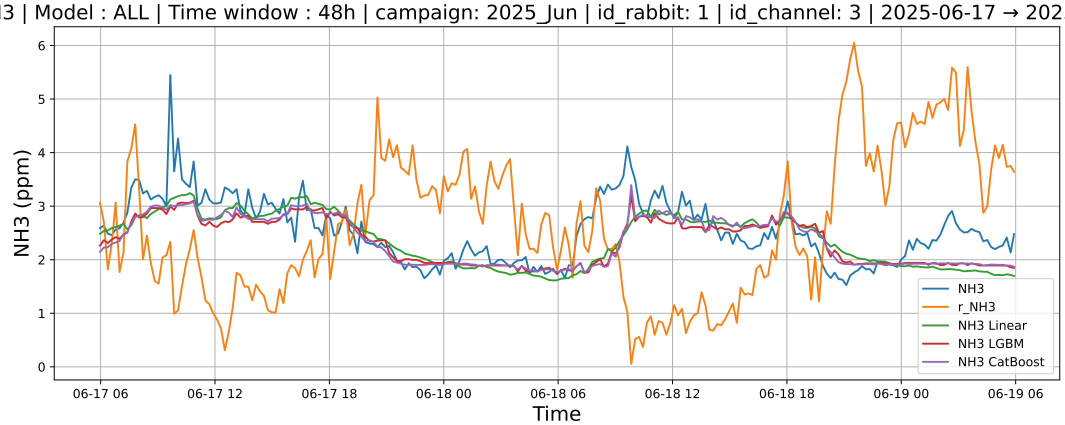
3.2.1.1 Time window : 24

NH3 | Model : ALL | Time window : 24h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-18 → 2025-06-19



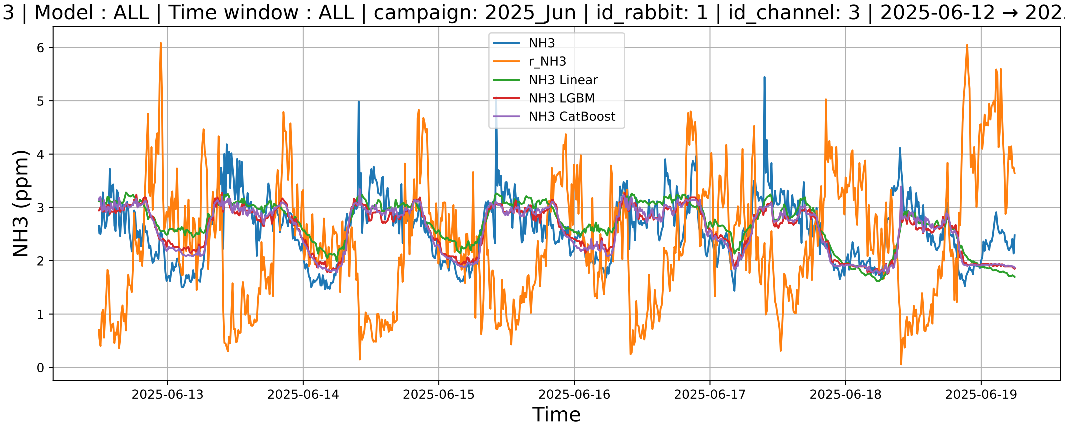
3.2.1.2 Time window : 48

NH3 | Model : ALL | Time window : 48h | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-17 → 2025-06-19



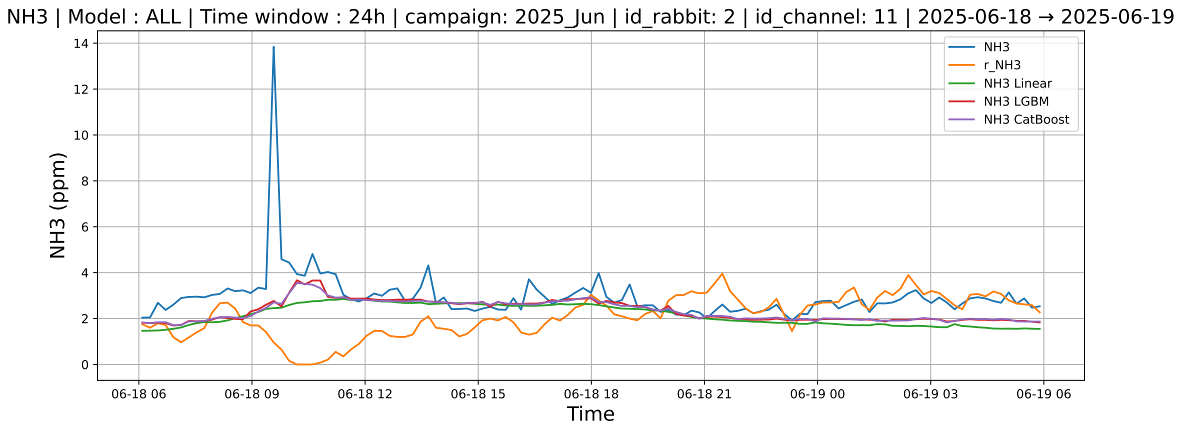
3.2.1.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Jun | id_rabbit: 1 | id_channel: 3 | 2025-06-12 → 2025-06-19

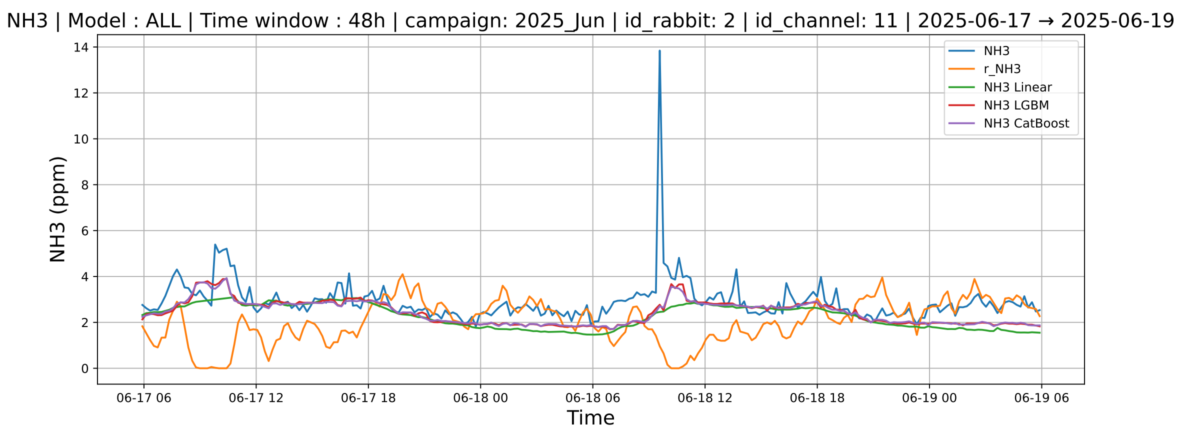


3.2.2 Group: campaign: 2025_Jun,id_rabbit: 2,id_channel: 11

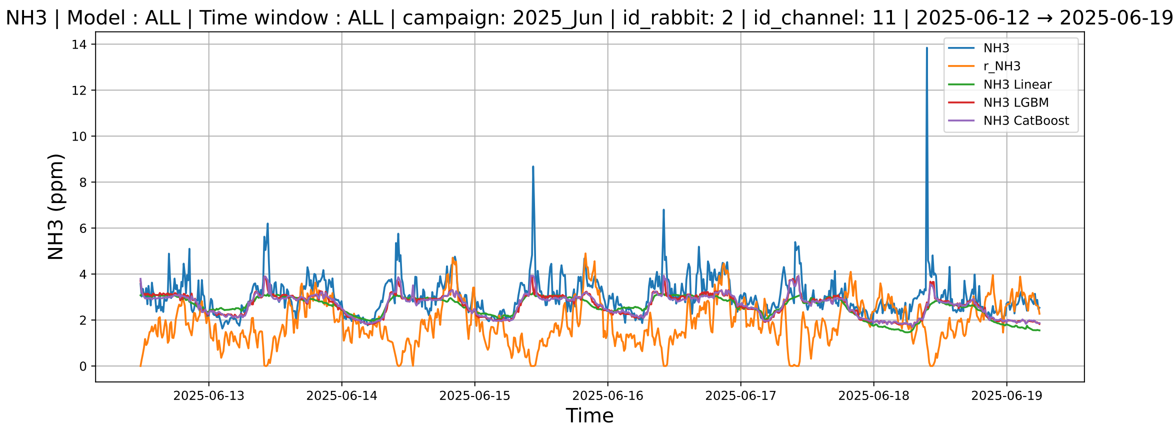
3.2.2.1 Time window : 24



3.2.2.2 Time window : 48

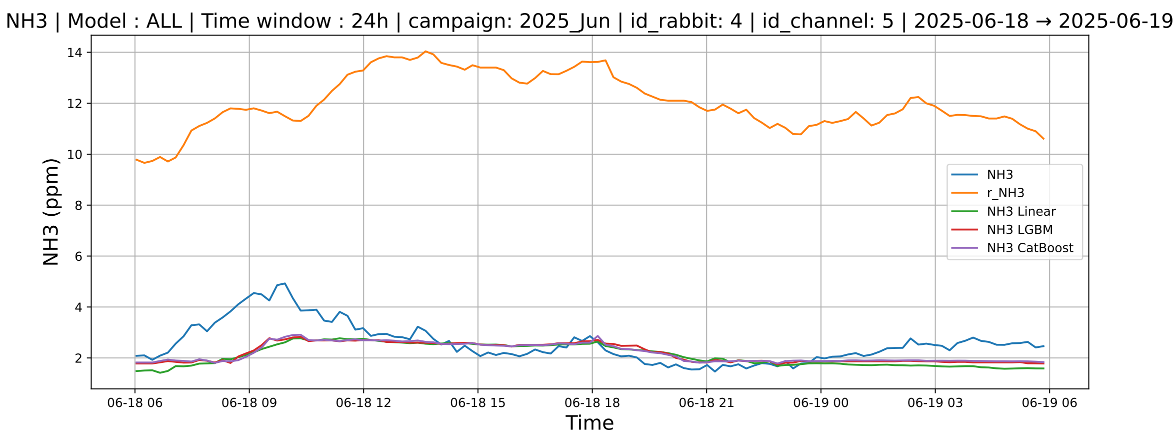


3.2.2.3 Time window : ALL

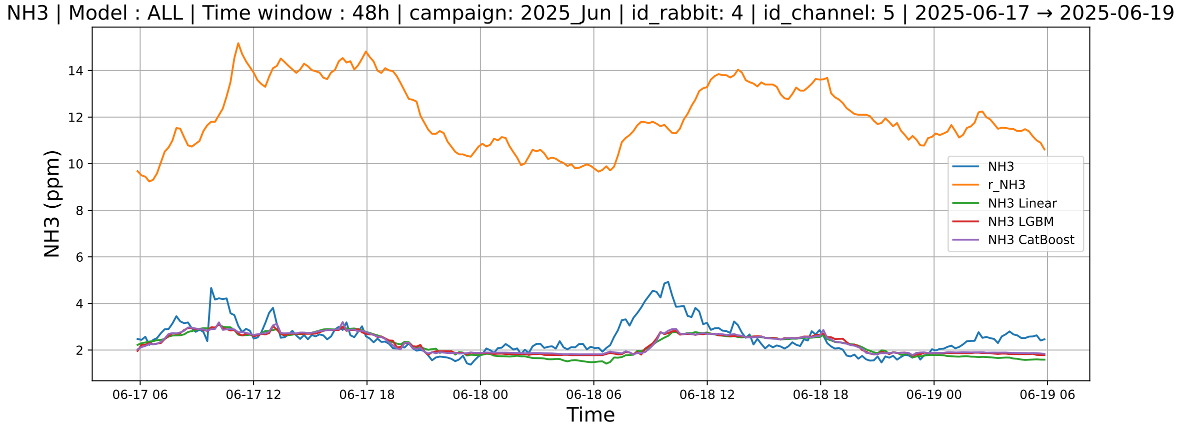


3.2.3 Group: campaign: 2025_Jun,id_rabbit: 4,id_channel: 5

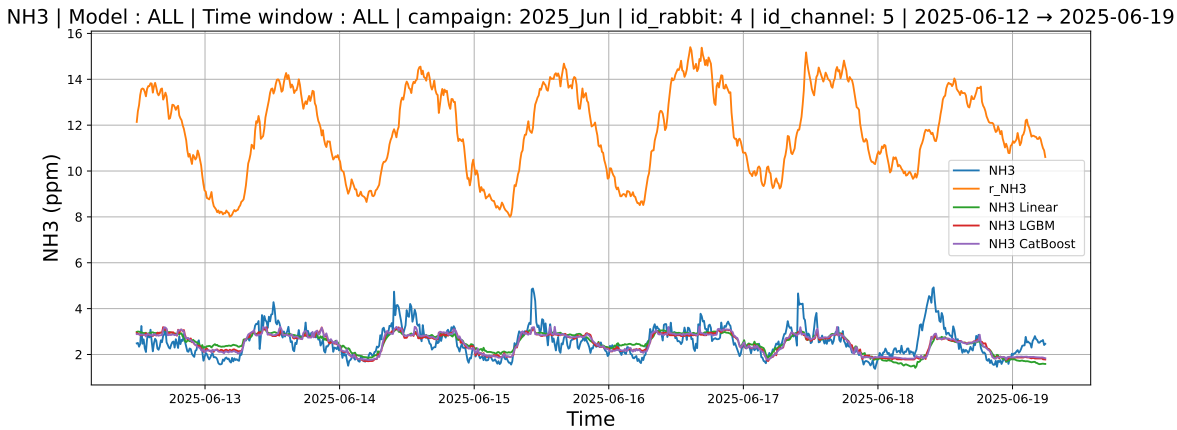
3.2.3.1 Time window : 24



3.2.3.2 Time window : 48

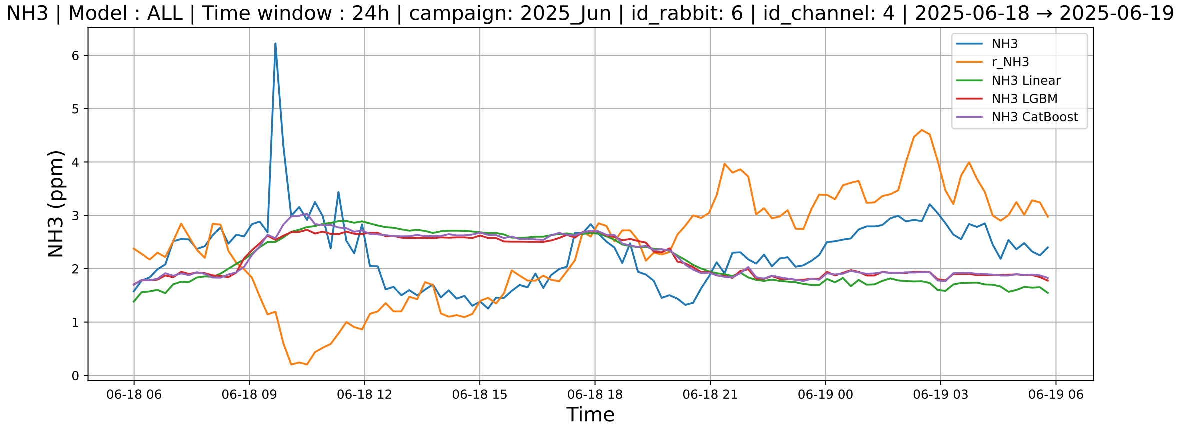


3.2.3.3 Time window : ALL

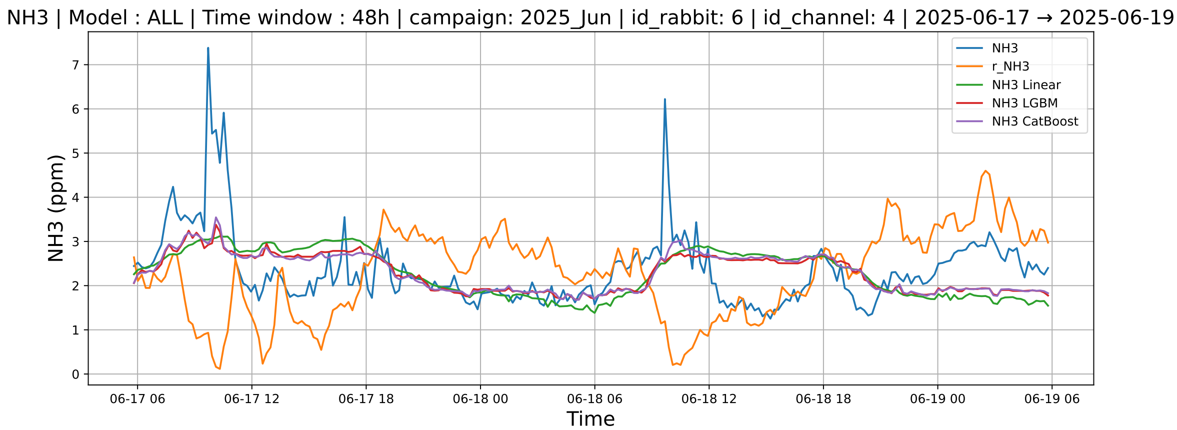


3.2.4 Group: campaign: 2025_Jun,id_rabbit: 6,id_channel: 4

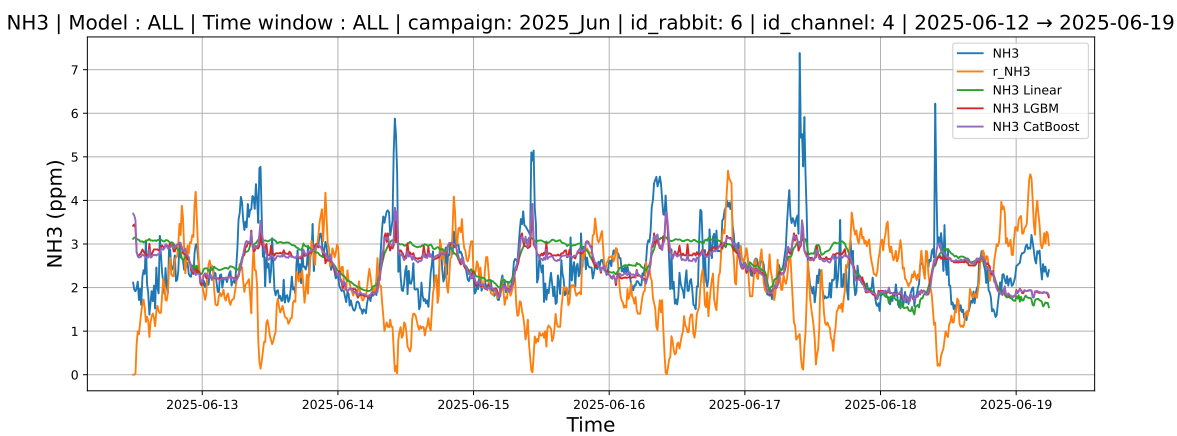
3.2.4.1 Time window : 24



3.2.4.2 Time window : 48

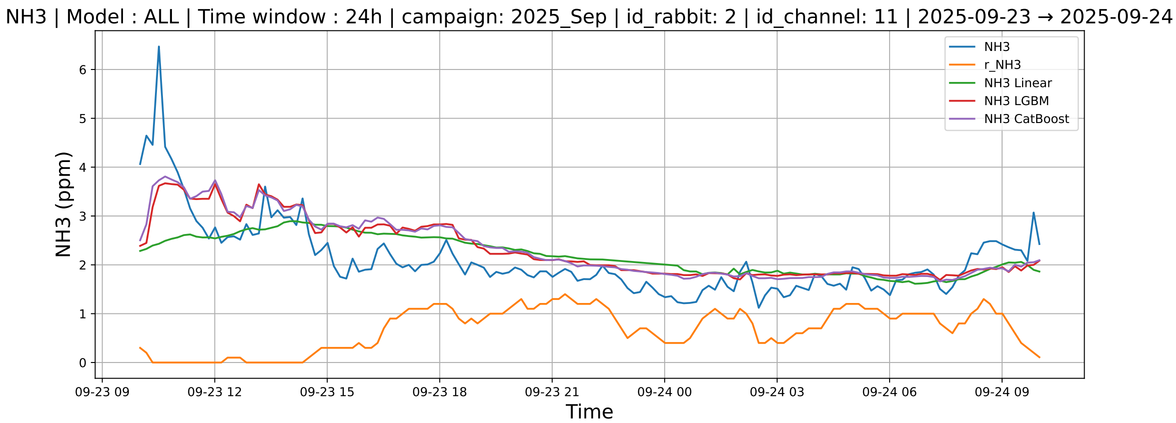


3.2.4.3 Time window : ALL

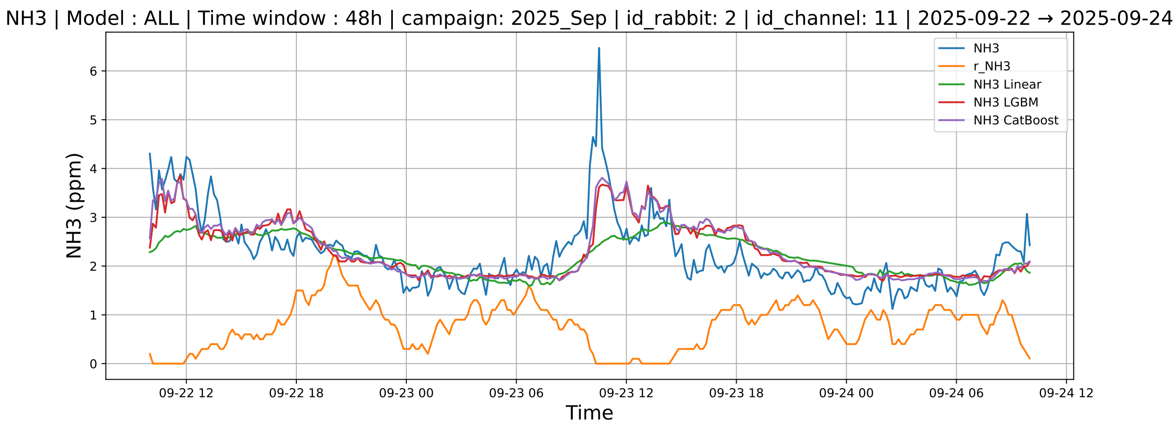


3.2.5 Group: campaign: 2025_Sep,id_rabbit: 2,id_channel: 11

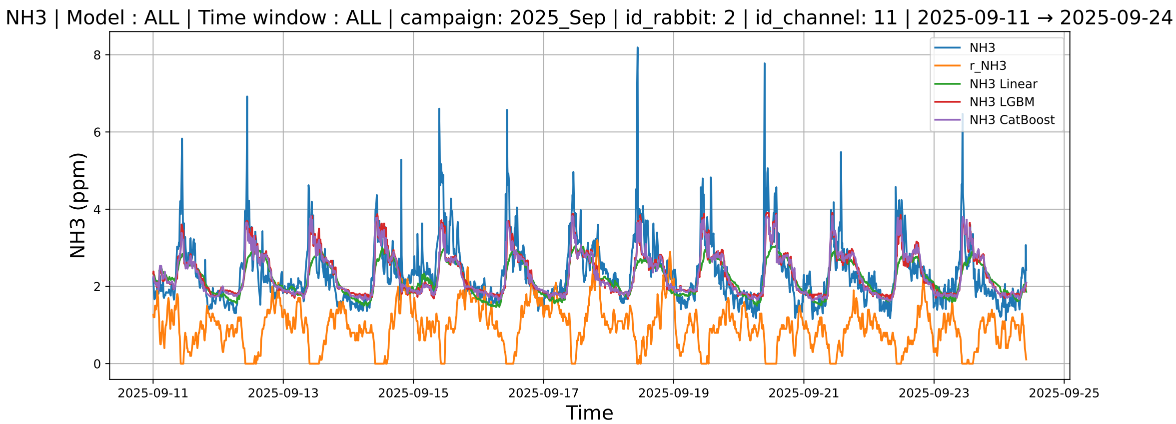
3.2.5.1 Time window : 24



3.2.5.2 Time window : 48

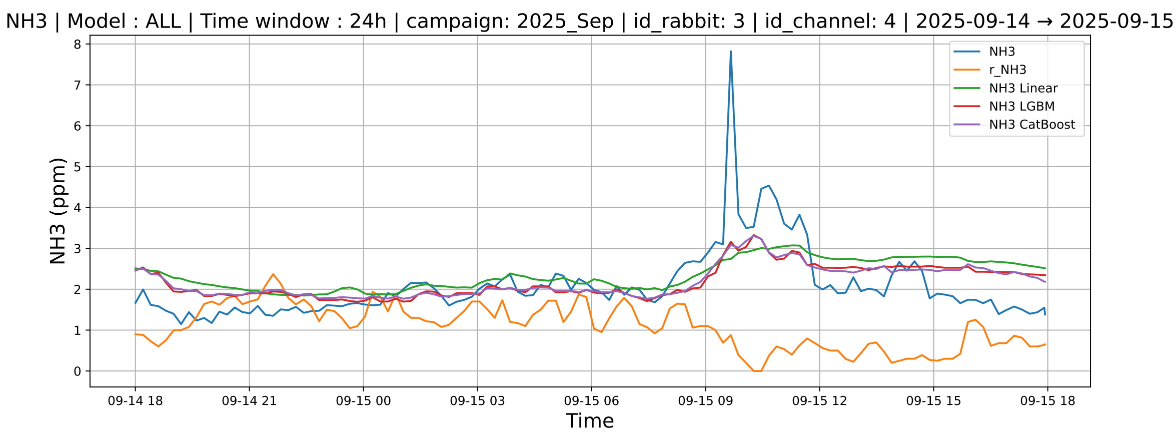


3.2.5.3 Time window : ALL

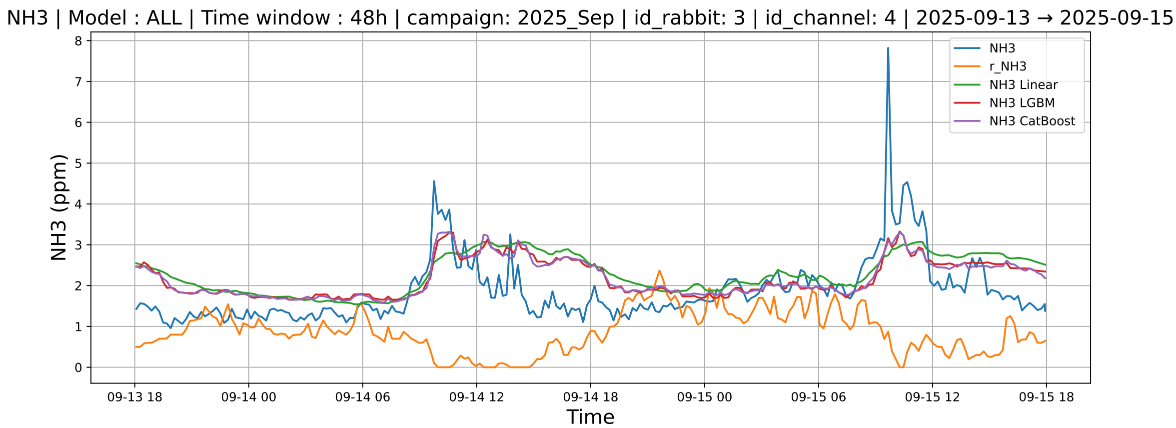


3.2.6 Group: campaign: 2025_Sep,id_rabbit: 3,id_channel: 4

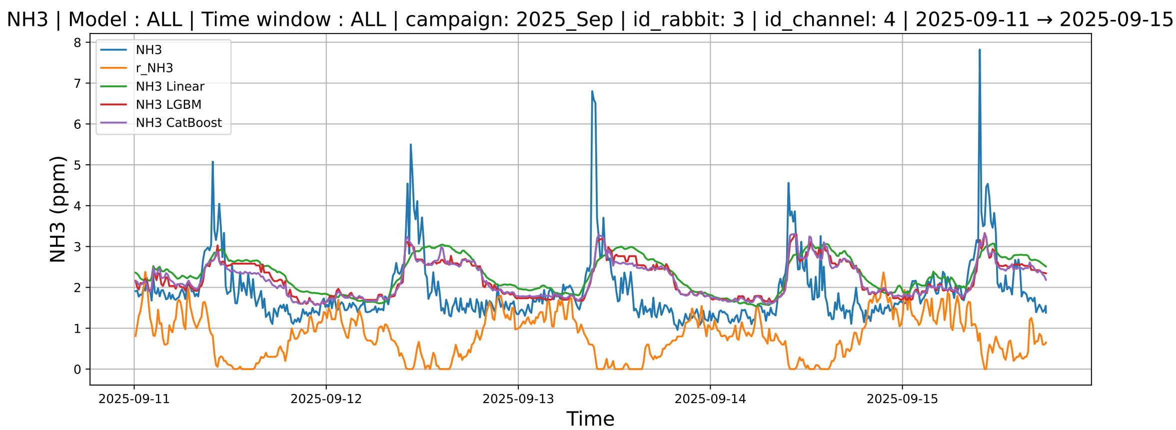
3.2.6.1 Time window : 24



3.2.6.2 Time window : 48

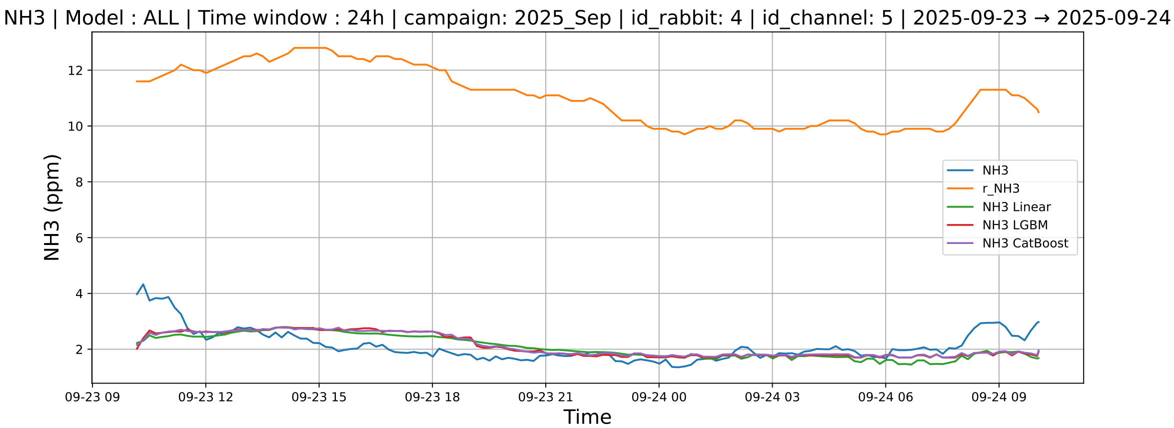


3.2.6.3 Time window : ALL

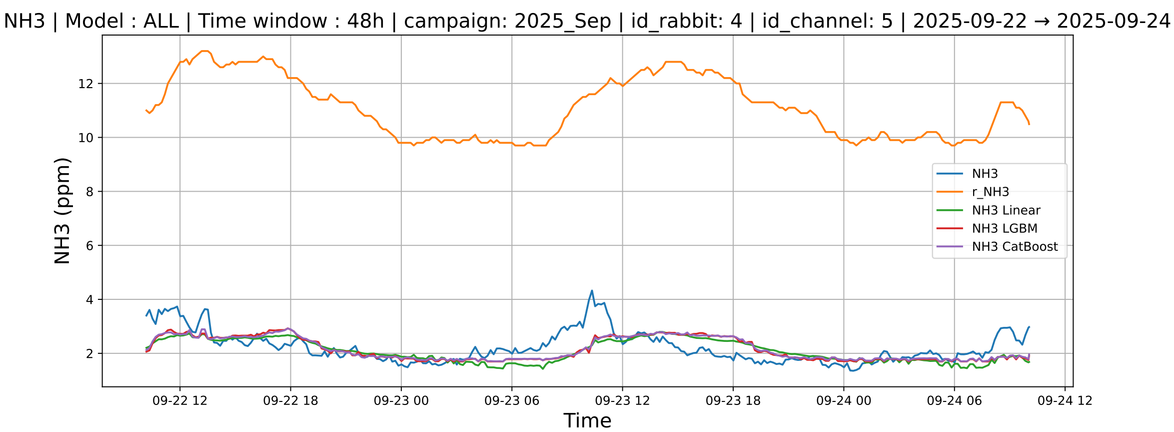


3.2.7 Group: campaign: 2025_Sep,id_rabbit: 4,id_channel: 5

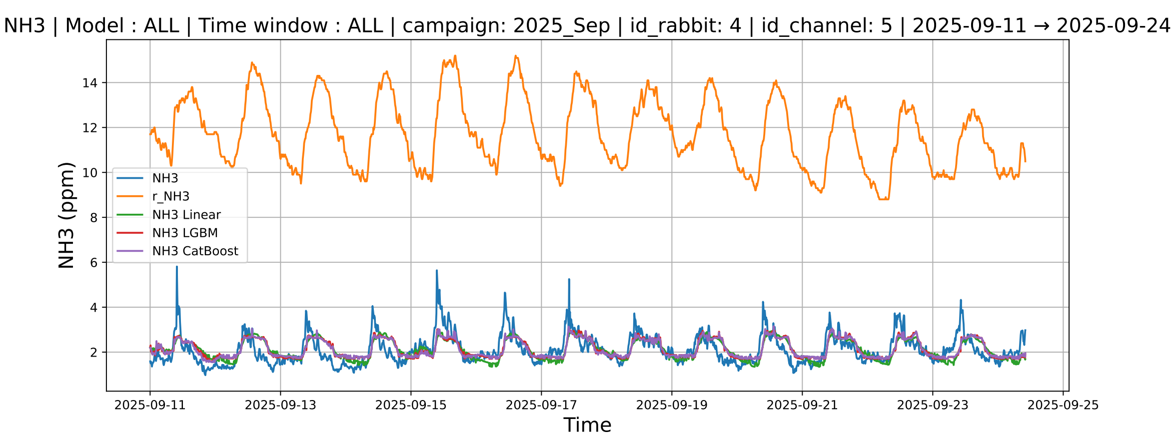
3.2.7.1 Time window : 24



3.2.7.2 Time window : 48

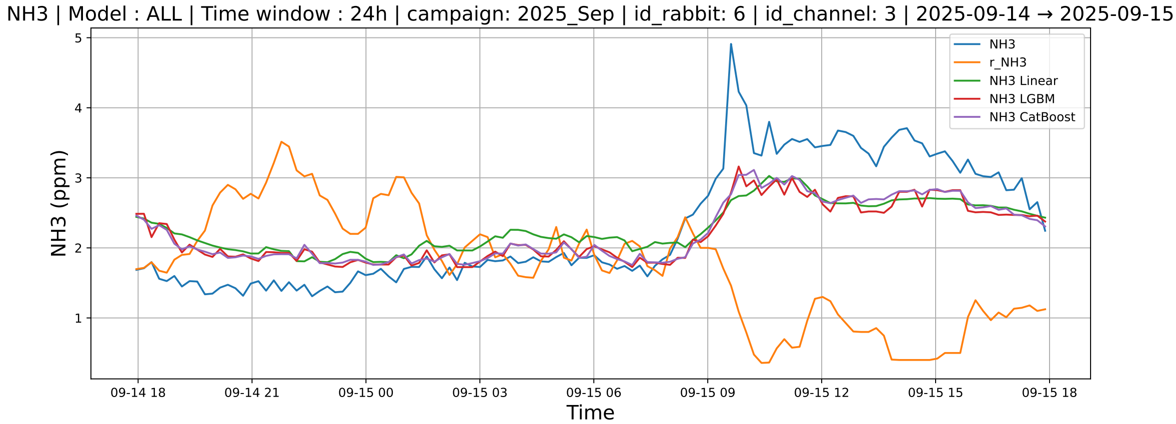


3.2.7.3 Time window : ALL

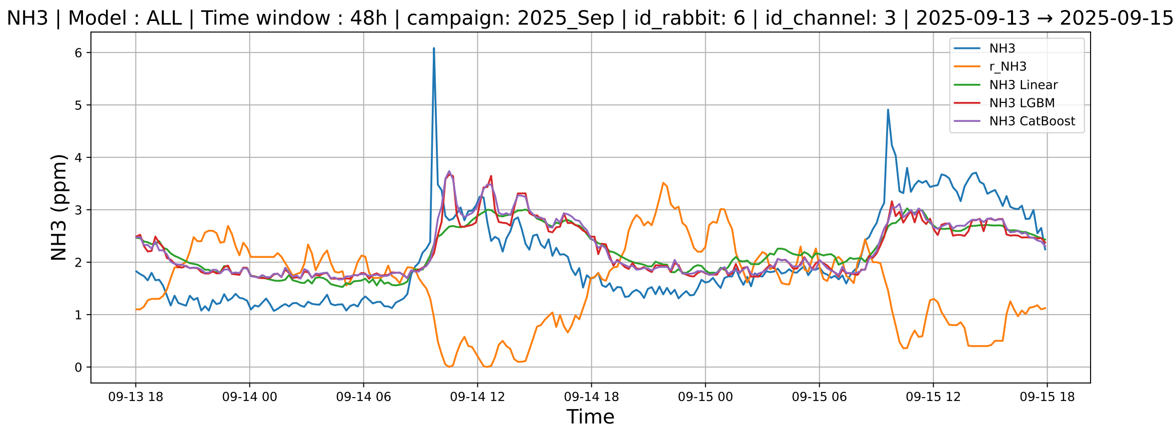


3.2.8 Group: campaign: 2025_Sep,id_rabbit: 6,id_channel: 3

3.2.8.1 Time window : 24



3.2.8.2 Time window : 48



3.2.8.3 Time window : ALL

NH3 | Model : ALL | Time window : ALL | campaign: 2025_Sep | id_rabbit: 6 | id_channel: 3 | 2025-09-11 → 2025-09-15